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Abstract

This paper studies the optimal asset allocation problem for an investor through utility maximization. A power utility function is adopted for this sake, and the model takes into account, taxes, and dividends and transaction costs. The assets available in the market are assumed to be risky asset, whose price follows a geometric Brownian motion, and riskless asset, given by the money market account. Interest rates are deterministic, and increase linearly over time with a slope equal to half the volatility of the risky asset. Transaction costs and Taxes are assumed to be proportional to the whole investment in the risky asset.

Understanding Power Utility Functions

Let w denote the investor's total wealth at the end of an investment period. A power utility function can be defined as:

$$U(w) = \begin{cases} \frac{w^{1-\phi}}{1-\phi}, & \phi \neq 1 \\ \ln(w), & \phi = 1 \end{cases}$$

where $\phi > 0$ is the coefficient of relative risk aversion. This function measures the investor's satisfaction or utility derived from final wealth w. It captures the principle of diminishing marginal utility: each additional unit of wealth provides less incremental satisfaction.

Note: Adding the constant term $-k$ to the utility function shifts it vertically downward. This does not affect the investor's preferences or optimal decisions, since it does not depend on wealth w; it simply re-anchors the utility scale.

To illustrate the interpretation, consider three types of investors, each characterized by a different value of ϕ. Suppose the investor's final wealth takes one of the following values: $w = 1$, $w = 2$, or $w = 10$. These may represent final outcomes such as \$1 million, \$2 million, or \$10 million, respectively.

For a risk-neutral investor with $\phi = 0$, the utility function simplifies to $U(w) = w$. The investor's utility increases linearly with wealth. The utility values corresponding to the three wealth levels are 1, 2, and 10, respectively. This investor values each additional dollar equally and is indifferent to risk.

For a mildly risk-averse investor with $\phi = 0.5$, the utility becomes $U(w) = 2\sqrt{w}$. In this case, the utility values for wealth levels $w = 1$, 2, and 10 are 2.00, approximately 2.83, and approximately 6.32, respectively. Utility increases as wealth increases, but the rate of increase diminishes. This reflects a cautious attitude: the investor values gains, but each additional dollar provides less added satisfaction than the previous one.

In contrast, for a highly risk-averse investor with $\phi = 2$, the utility function becomes $U(w) = -\frac{1}{w}$. The corresponding utility values are -1.00, -0.50, and -0.10 for the same wealth levels. Although the utility values are negative, they become closer to zero as wealth increases, indicating increasing satisfaction. However, the growth in utility is even slower than in the mildly risk-averse case. This reflects stronger risk aversion: the investor gains little additional satisfaction from higher wealth and is highly averse to financial loss.

These examples demonstrate how the power utility function encodes different attitudes toward risk. The higher the value of ϕ, the more risk-averse the investor is, and the more slowly utility increases with wealth. In all cases, the utility function serves as a formal tool for evaluating the desirability of different investment outcomes based on the investor's preferences.

Transaction Costs

Transaction costs and taxes are assumed to be proportional to the total amount invested in the risky asset. This means that as the size of the investment in the risky asset increases, the total cost due to taxes and transaction fees increases proportionally.

Mathematically, if x denotes the amount invested in the risky asset, the total cost incurred is given by:

$$\text{Cost} = \tau x + \nu x = (\tau + \nu)x,$$

Introduction

Optimal portfolio allocations are derived when an investor maximizes the utility of his wealth. The portfolio problem is a problem involving choice. The choice set typically contains elements such as how much to invest, what to invest in, what point in one's life-cycle one should invest in what assets, how should one invest in order to have maximum tax benefits over the life-cycle, etc. An investor saves from current earnings for future consumption and invests the savings in investment vehicles, broadly classified into asset classes like stocks, bonds, gold, real estate, etc. The investor evaluates the performance of the investments using some criterion; this is called the investor's objective. Let us denote the investor's objective by a function, U(.); this can be a function of returns or wealth or any other aspect of the investment the investor values. Moreover, since the future returns of the asset classes is uncertain, the wealth one will have at future times is uncertain; in this case the objective is taken to be the expected value of the function, U(.).Earlier work in this area on optimal portfolio selection problem can be traced to Markowitz's mean variance model (Markowitz, 1959); Samuelson (Sameulson, 1969) extended the work of Markowitz to a dynamic set up. He used a dynamic stochastic programming approach; he succeeded in obtaining the optimal decision for consumption investment model. Merton (Merton, 1971) used the stochastic optimal control methods in continuous finance to obtain a closed form solution to the problem of optimal portfolio strategy under specific assumptions about the asset returns and the investor's preferences. These days, investors invest both in the money market and stocks. Due to the high risks involved in the stock market, investment strategies and risk management are becoming more important. Using the Hamilton-Jacobi-Bellman equation, (Hipp et al., 2000) determined the strategy of investment which minimizes the probability of ruin modeling the price of the stocks by geometric Brownian motion. Gaier et al. (Gaier, et al., 2002) under the same hypothesis obtained an exponential bound with a rate that improves the classical

Investing Background Information

An investor typically saves a portion of their current income rather than spending it all immediately. These savings are intended for future consumption, such as retirement, major purchases, or unexpected expenses.

Instead of holding the money passively, the investor allocates it to investment vehicles with the aim of generating returns over time. These investment vehicles are broadly categorized into asset classes, which include stocks (equity ownership in companies), bonds (debt instruments issued by corporations or governments), commodities like gold, and real estate (property and land). Each asset class offers different risk and return profiles, and the investor's allocation among them depends on their financial goals, investment horizon, and risk tolerance.

Note: Hipp et al. (2000) used advanced mathematical techniques—specifically, the Hamilton-Jacobi-Bellman (HJB) equation—to determine the optimal investment strategy in a risky asset whose price evolves randomly, following a stochastic process (geometric Brownian motion). The goal of their analysis was to minimize the probability of ruin, that is, to reduce the likelihood of the investor going bankrupt.

Exponential Bounds and the Improvement over the Lundberg Parameter

Gaier et al. (2002), working under the same modeling assumptions as Hipp et al. (2000), derived an improved exponential bound on the probability of ruin. Specifically, they considered a framework in which stock prices follow a geometric Brownian motion and investment strategies are evaluated using the Hamilton-Jacobi-Bellman (HJB) equation. The central objective remained the same: to minimize the probability that an investor or insurer is ruined—i.e., runs out of money—over time.

In this context, an *exponential bound* refers to an inequality that provides an upper limit on the probability of ruin, and this upper limit decreases exponentially fast as the initial capital increases. Mathematically, such a bound takes the form:

$$\mathbb{P}(\text{ruin}) \leq Ce^{-\lambda x},$$

where x is the initial capital, C is a positive constant, and λ is the exponential rate of decay. This bound implies that as x becomes larger, the probability of ruin becomes negligibly small at an exponential rate. For example, if a person starts with \$1,000, their chance of ruin might be bounded by 5%; but starting with \$2,000 reduces that upper bound to 1%, and starting with \$3,000 lowers it further to below 0.2%. Each additional unit of capital yields a rapidly diminishing risk of ruin, offering increasing financial security.

In classical ruin theory, this type of exponential behavior is often characterized by the *Lundberg parameter*, which serves as the exponent in the classical exponential bound:

$$\psi(x) \approx e^{-\theta x},$$

where $\psi(x)$ denotes the probability of ruin given initial capital x, and θ is the Lundberg exponent. A larger value of θ implies a faster decay of ruin probability with respect to capital.

Gaier et al. demonstrated that under their model, they could obtain an exponential bound with a rate of decay λ that exceeds the classical Lundberg parameter θ. This means their bound predicts a more rapid decline in ruin probability as capital increases, thereby improving upon traditional models. Their result offers both a tighter risk assessment and practical implications for setting safer capital reserves. Their strategy involved investing a fixed monetary amount in the risky asset, regardless of the investor's total reserve. This implies that the optimal allocation to the stock remained constant in absolute terms and did not vary with changes in the investor's overall wealth.

Castillo and Parrocha (Castillo et al., 2008) considered an insurance business with a fixed amount available for investment in a portfolio consisting of one non-risky asset and one risky asset. They presented the Hamilton-Jacobi-Bellman (HJB) equation and demonstrated its use in finding the optimal investment strategy based on some given criteria. The objective of the resulting control problem was to determine the investment strategy that minimized infinite ruin probability. The existence of a solution to the resulting HJB equation was then shown by verification theorem. A numerical algorithm is also given for analysis. Promislow and Young (Promislow et al., 2005) minimized the probability of ruin of an insurer facing a claim process modeled by a Brownian motion with drift. They consider two controls to minimize the probability of ruin;

Castillo and Parrocha

Castillo and Parrocha (2008) studied an optimal investment problem for an insurance company that has a fixed amount of capital to allocate between a non-risky asset and a risky asset. Their goal was to determine how the insurer should invest this capital in order to minimize the probability of ruin over an infinite time horizon. To address this, they used a mathematical modeling approach known as a *stochastic control framework*, which involves making decisions over time in an environment where outcomes are uncertain and influenced by random factors—such as the unpredictable returns of a risky asset. Within this framework, they formulated and solved the Hamilton-Jacobi-Bellman (HJB) equation, which characterizes the value of the optimal strategy at any point in time. To ensure that their proposed solution to the HJB equation was truly optimal, they applied the *verification theorem*, a mathematical result that confirms whether a candidate solution satisfies all the necessary conditions to be considered the best possible strategy. In addition to these theoretical results, they provided a numerical algorithm to compute the solution, offering a practical method for insurers to evaluate and implement strategies that minimize the long-term risk of insolvency.

1. Investing in a risky asset (constrained and the non-constrained cases)

2. Purchasing quota-share reinsurance.

They obtained an analytic expression for the minimum probability of ruin and their corresponding optimal controls. They also demonstrated their results with numerical results.

Promislow and Young

Promislow and Young (2005) investigated the problem of minimizing the probability of ruin for an insurance company whose liabilities are driven by a continuous claim process modeled as a Brownian motion with drift. In this setting, claims arrive randomly and continuously over time, but with an overall upward trend (drift). To mitigate the risk of insolvency, the authors introduced two control mechanisms: investment in a risky asset, and the purchase of quota-share reinsurance. They examined both constrained and unconstrained investment scenarios, where the constrained case imposes limits on how much can be invested in the risky asset, while the unconstrained case allows unrestricted investment decisions, including leverage and short-selling. By applying techniques from stochastic control theory, they derived an explicit analytic expression for the minimum probability of ruin and identified the corresponding optimal strategies for both controls. Their theoretical findings were further validated through numerical simulations, demonstrating the practical relevance of their results for managing long-term financial risk in insurance operations.

In (Bayraktar et al., 2008), a minimization problem using stochastic optimal control techniques with the assumption that an agent's rate of consumption is ratcheted; that is it forms a non-decreasing process, was solved. Liu and Yang (Liu et al., 2004) studied optimal investment strategies of an insurance company where they assumed that an insurance company receives premiums at a constant rate. The total claims are modeled by a compound Poison process, and the insurance company can invest in the money market (bonds) and in a risky asset such as stocks. In (Oksendal et al., 2002), a market with one risk-free and one risky asset in which the dynamics of the risky asset are governed by a geometric Brownian motion was investigated with the objective to maximize the cumulative expected utility of consumption over planning horizon. Kostadinova (Kostadinova, 2007) considered a stochastic model for the wealth of an insurance company which has the possibility to invest into a risky asset and a risk-less asset under constant mix strategy. Here the resulting integrated risk process and the corresponding discounted net loss process were investigated. This opened up a way to measure the risk of a negative outcome of the integrated risk process in a stationary way. It also provided an approximation of the optimal investment strategy that maximizes the expected wealth of the insurance company under the risk constraint on the Value-at-Risk.

Bayraktar

Bayraktar et al. (2008) addressed a minimization problem within a stochastic optimal control framework, where the objective was to reduce the probability of ruin or some other measure of financial risk. A distinctive feature of their model was the assumption that the agent's rate of consumption is *ratcheted*, meaning it is a non-decreasing process over time. Formally, this constraint is expressed as

$$c(t_2) \geq c(t_1) \quad \text{for all } t_2 > t_1,$$

where $c(t)$ denotes the consumption rate at time t. Once the agent increases their level of consumption, they are not permitted to reduce it in the future. This condition reflects a realistic behavioral tendency observed in financial decision-making, where individuals are often unwilling or unable to lower their standard of living after it has risen. Mathematically, this introduces additional complexity into the control problem, as the investor must manage their wealth and investment strategy not only to avoid ruin, but also to sustain a permanently increasing level of consumption over time.

Key Definitions

Insurance Premium: An *insurance premium* is the fixed amount paid by policyholders to an insurance company in exchange for coverage against specified risks. In modeling terms, the premium represents a deterministic inflow of cash that contributes to the insurer's reserve. When it is assumed that the insurer receives premiums at a constant rate, this implies a steady, time-continuous accumulation of income. Mathematically, if p denotes the constant premium rate, then the total premium income over time t is given by:

$$\text{Premium income} = p \cdot t.$$

Constant Mix Strategy: A *constant mix strategy* is a portfolio management rule where a fixed proportion of total wealth is continuously maintained in each asset class. Specifically, if $\pi \in [0, 1]$ denotes the fraction of wealth allocated to a risky asset (such as stocks), then $(1 - \pi)$ is invested in a risk-free asset (such as bonds). As asset values fluctuate over time, the portfolio is rebalanced to preserve the original allocation. This strategy is designed to maintain consistent risk exposure and often results in selling outperforming assets and buying underperforming ones. At any time t, the investment amounts are given by:

$$\text{Investment in risky asset} = \pi \cdot W_t, \qquad \text{Investment in risk-free asset} = (1 - \pi) \cdot W_t,$$

where W_t denotes the total wealth at time t.

Kostadinova

Kostadinova (2007) studied how an insurance company's wealth evolves over time when the company invests in both a risky asset and a risk-free asset while following a constant mix strategy. In her analysis, she focused on two key processes. The first is the *integrated risk process*, which refers to the total accumulation of financial risk over time, such as the sum of all claims made against the company or its total exposure to uncertain outcomes. The second is the *discounted net loss process*, which measures how much total loss the company might experience, but adjusts that loss to reflect its present-day value—this accounts for the fact that money lost in the future is worth less than money lost today. Kostadinova analyzed both of these processes within what is called a *stationary framework*, meaning that the long-term statistical behavior of the risk—such as its mean or variance remains stable over time. This allows the company to assess and manage its long-run risk in a predictable and consistent way. In addition to analyzing risk, she also studied how to increase the company's expected wealth. However, this growth objective was subject to a constraint: the company's investment strategy could not exceed a certain level of risk as measured by *Value-at-Risk (VaR)*. Value-at-Risk is a standard risk metric in finance that estimates the worst loss a company might face over a fixed time period, with a given level of confidence. For example, it might state that there is only a 5% chance that losses will exceed a specific dollar amount over the next year. Using this constraint, Kostadinova proposed an approximate investment strategy that aims to grow the company's wealth as much as possible while ensuring that the chance of severe financial loss remains acceptably low.

Liu, Bai and Yiu (Bai et al., 2007) considered a constrained investment problem with the objective of minimizing the ruin probability and formulated the cash reserve and investment model for the insurance company and analyzed the Value-at-Risk in a short time horizon.

In this paper however, optimal investment problem of utility maximization with taxes, dividends and transaction costs under power Utility function is studied. With the application of the Ito lemma, we obtained the Hamilton-Jacobi-Bellman equation associated with the optimization problem for the power utility maximization of the investor. Furthermore, the properties of the optimal strategies are analyzed and the effects of market parameters on the strategies discussed.

This work aims at optimization of an investor's investment returns, build a foundation for further research and also contribute to improved business ventures.

- They use Ito's lemma, a tool from stochastic calculus, to handle the randomness in asset prices (modeled as stochastic processes).
- Applying Ito's lemma lets them derive the Hamilton-Jacobi-Bellman (HJB) equation, which is a differential equation that characterizes the value function — i.e., the maximum expected utility the investor can achieve at any point in time.
- Solving this HJB equation gives them the optimal investment strategy: how much to put in risky vs. risk-free assets, given taxes, transaction costs, and dividends.

Understanding the Stochastic Differential Equation

The equation

$$dX_t = \mu_t\, dt + \sigma_t\, dW_t$$

is a stochastic differential equation that describes how a variable X_t evolves over time when both predictable trends and random fluctuations are present. The variable X_t represents a stochastic process—such as a stock price, an investor's wealth, or the value of a portfolio—that changes with time and includes randomness. The term dX_t denotes the infinitesimal change in X_t over an instant of time. The component $\mu_t\, dt$ is called the drift term; it represents the smooth, average direction in which X_t tends to move, where μ_t is the expected rate of change and dt is a very small time increment. The component $\sigma_t\, dW_t$ is the diffusion term, where σ_t is the volatility (how sensitive the process is to randomness), and dW_t is an increment of Brownian motion—a mathematical model for random, continuous fluctuations. This term accounts for the unpredictable, noisy part of X_t's movement. Taken together, the equation states that the process evolves through a combination of a deterministic drift and a random shock. This type of model is fundamental in financial mathematics, where asset prices and investment strategies are frequently influenced by both systematic trends and market uncertainty.

Understanding Itô's Lemma from a Calculus Perspective

Let $f(x,t) = x^2 t$. This is a concrete function that takes two inputs: a number x and a time t, and outputs their product with x squared. In standard multivariable calculus, if both x and t are changing, then the total derivative of f is given by the chain rule:

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dx.$$

Here, $\frac{\partial f}{\partial t} = x^2$, and $\frac{\partial f}{\partial x} = 2xt$, so we get:

$$df = x^2\, dt + 2xt\, dx.$$

Applying Itô's Lemma Step by Step: Example with $f(x,t) = x^2 t$

Let us consider a specific function:

$$f(x,t) = x^2 t,$$

which depends on two variables: a number x, and time t. Now we want to study what happens if x is not just a constant or smooth input, but a random process that evolves over time. Specifically, let $x = X_t$, where X_t follows the stochastic differential equation:

$$dX_t = \mu_t dt + \sigma_t dW_t.$$

Here:

- X_t is a stochastic process — a variable that moves randomly as time passes.
- μ_t is called the drift — the average rate at which X_t changes over time.
- σ_t is the volatility — it determines how strongly X_t responds to random noise.
- dW_t is the increment of a Brownian motion — a model for pure, continuous-time randomness.

We now want to understand how the function $f(X_t, t) = X_t^2 t$ changes over time as X_t moves randomly. That means we are computing the differential df, just like we do with the chain rule in multivariable calculus — except here we need to use Itô's Lemma, because X_t is random.

Itô's Lemma tells us how to compute df when f depends on a stochastic process X_t and time t. The formula is:

$$df = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2}\sigma_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dW_t.$$

We now compute each partial derivative:

$$\frac{\partial f}{\partial t} = x^2, \quad \frac{\partial f}{\partial x} = 2xt, \quad \frac{\partial^2 f}{\partial x^2} = 2t.$$

Now plug these into the formula:

$$df = \left(x^2 + \mu_t \cdot 2xt + \frac{1}{2}\sigma_t^2 \cdot 2t \right) dt + \sigma_t \cdot 2xt \cdot dW_t.$$

Simplify:

$$df = \left(x^2 + 2xt\mu_t + \sigma_t^2 t\right) dt + 2xt\sigma_t dW_t.$$

This is the full differential of the function $f(X_t, t) = X_t^2 t$. The first part (the dt term) gives the average or expected change, and the second part (the dW_t term) captures how randomness affects f at each moment. The second derivative term, $\frac{1}{2}\sigma_t^2 \frac{\partial^2 f}{\partial x^2}$, appears only because X_t includes randomness — that's what makes this different from the normal chain rule.

Note: Itô's Lemma only applies when the variable X_t follows a stochastic differential equation (SDE) of the specific form:

$$dX_t = \mu_t \, dt + \sigma_t \, dW_t,$$

which defines what is known as an *Itô process*.

The Problem Formulation

> **We shall assume that the investor can trade two assets continuously in an economy. The first asset is the money market account (bond) growing at a rate r_t that is linear function of time ($r_t = \alpha + \sigma t$),**

We are given that the interest rate r_t on the money market account (or bond) evolves according to the linear equation

$$r_t = \alpha + \sigma t,$$

where α represents the initial interest rate at time $t = 0$, and σ denotes the rate at which the interest rate increases over time. This specification implies that the bond does not yield a constant return, but rather earns interest at a rate that grows linearly with time. As a result, the longer the bond is held, the higher the interest rate becomes. For example, at time $t = 0$, the rate is α; at time $t = 1$, the rate is $\alpha + \sigma$; and at time $t = 2$, it is $\alpha + 2\sigma$, and so forth. This formulation introduces time-dependent dynamics into the model, making it more flexible and capable of capturing gradually increasing interest rate environments.

> **instead of a constant as in (Nie, 2010). $r_t = \alpha + \sigma t$, *($\alpha > 0, 0 \leq \sigma < \infty$) is a decreasing (or an increasing) linear* function of t as $t \to \infty$. If $\sigma > 0$, then r_t is increasing and constant if $\sigma = 0$. This property is now valid for any t not only when $t \to \infty$. Hence r_t is simply a deterministic linear process and σ is the slope and not just the acceleration coefficient which is the volatility (variance) of the process and is proportional to the level of the interest rate. If the rate is assumed to follow the Ornstein-Uhlenbeck process**
>
> $dr_t = \alpha(\beta - r_t)dt + \sigma dZ(t), \quad r_{t_0} = r_0.$
>
> **Then**
>
> $r_t = (r_0 - \beta)e^{-\alpha t} + \sigma \int_{t_0}^{t} e^{-\alpha(t-u)} dZ(u),$

Clarifying the Role of σ in the Deterministic Linear Interest Rate Model

Hence r_t is simply a deterministic linear process and σ is the slope and not just the acceleration coefficient which is the volatility (variance) of the process and is proportional to the level of the interest rate.

In the model $r_t = \alpha + \sigma t$, the interest rate r_t evolves in a predictable, non-random way. This defines a deterministic linear process, meaning there is no stochastic component involved. The parameter σ controls how fast r_t increases (or decreases) over time; it represents the slope of the line in the same way that m is the slope in the linear equation $y = mx + b$.

The original phrasing describing σ as "not just the acceleration coefficient which is the volatility" is confusing. It appears to draw an analogy to stochastic models, such as those involving Brownian motion, where σ does represent volatility. However, in this deterministic model, σ does not describe randomness or variation—it simply sets a constant trend.

Finally, the statement that σ is "proportional to the level of the interest rate" is misleading. In the equation $r_t = \alpha + \sigma t$, the slope σ is constant and does not change with r_t. While r_t increases over time, this increase is linear, and σ remains unchanged throughout.

The Ornstein–Uhlenbeck Process

The Ornstein–Uhlenbeck process models the evolution of the interest rate r_t through the stochastic differential equation

$$dr_t = \alpha(\beta - r_t)\,dt + \sigma\,dZ_t.$$

In this equation, dt represents an infinitesimally small increment of time, and dr_t denotes the change in the interest rate over that interval. The first term, $\alpha(\beta - r_t)dt$, governs the deterministic component of the rate's evolution. It encodes the idea of mean reversion: if the current rate r_t is below the long-run average β, the term $\beta - r_t$ is positive and the product $\alpha(\beta - r_t)dt$ nudges the rate upward; conversely, if r_t exceeds β, the term becomes negative and pulls the rate downward. The parameter $\alpha > 0$ controls the speed of this reversion. Importantly, the correction is not instantaneous but proportional to both the deviation from the mean and the small time increment dt. Over time, this results in a gradual convergence toward the average level β. The second term, $\sigma\,dZ_t$, introduces randomness via dZ_t, a Brownian motion increment, scaled by the volatility parameter σ. This ensures that the process exhibits stochastic fluctuations around the mean-reverting path. Together, the two terms allow the interest rate to move randomly while remaining anchored to a long-run equilibrium, making the Ornstein–Uhlenbeck process a realistic model for interest rate dynamics in continuous time.

Solving the Ornstein–Uhlenbeck SDE Using the Itô Product Rule

We begin with the stochastic differential equation:

$$dr_t = \alpha(\beta - r_t)\,dt + \sigma\,dZ(t).$$

This is a linear stochastic differential equation with respect to r_t, which represents the interest rate at time t. To solve it, we apply the method of integrating factors.

Define the integrating factor:

$$M(t) = e^{\alpha t}.$$

Multiply both sides of the equation by $M(t)$:

$$M(t)\,dr_t = M(t)\,\alpha(\beta - r_t)\,dt + M(t)\,\sigma\,dZ(t).$$

Apply **Itô's product rule** to the product $M(t)\,r_t$:

$$d(M(t)\,r_t) = M(t)\,dr_t + r_t\,dM(t).$$

Compute $dM(t)$ using the chain rule:

$$dM(t) = \frac{d}{dt}e^{\alpha t}\,dt = \alpha e^{\alpha t}\,dt = \alpha M(t)\,dt.$$

Substitute $dM(t)$ into the product rule:

$$d(M(t)\,r_t) = M(t)\,dr_t + \alpha r_t M(t)\,dt.$$

Factor out $M(t)$ from the right-hand side:

$$d(M(t)\, r_t) = M(t)\, [dr_t + \alpha r_t\, dt]\, .$$

Substitute the expression for dr_t from the original SDE:

$$dr_t = \alpha(\beta - r_t)\, dt + \sigma\, dZ(t),$$

So we get:

$$d(M(t)\, r_t) = M(t)\, [\alpha(\beta - r_t)\, dt + \sigma\, dZ(t) + \alpha r_t\, dt]\, .$$

Distribute and simplify:

$$d(M(t)\, r_t) = M(t)\, [\alpha\beta\, dt + \sigma\, dZ(t)]\, .$$

Simplifying:

$$d(M(t)r_t) = \alpha\beta e^{\alpha t} dt + \sigma e^{\alpha t} dZ(t).$$

We previously obtained:

$$d(M(t)\, r_t) = \alpha\beta M(t)\, dt + \sigma M(t)\, dZ(t).$$

Now we integrate both sides from t_0 to t.

Left-hand side:

Recall that when you integrate a differential $dF(t)$, you're summing up the infinitesimal changes in F.

$$\int_{t_0}^{t} d(M(u)\, r_u) = M(t)\, r_t - M(t_0)\, r_{t_0}.$$

Right-hand side:

We integrate each term:

$$\int_{t_0}^{t} \alpha\beta M(u)\, du + \int_{t_0}^{t} \sigma M(u)\, dZ(u).$$

Putting both sides together:

$$M(t)\, r_t - M(t_0)\, r_{t_0} = \alpha\beta \int_{t_0}^{t} M(u)\, du + \sigma \int_{t_0}^{t} M(u)\, dZ(u).$$

Now solve for $M(t)\, r_t$:

$$M(t)\, r_t = M(t_0)\, r_{t_0} + \alpha\beta \int_{t_0}^{t} M(u)\, du + \sigma \int_{t_0}^{t} M(u)\, dZ(u).$$

Since $M(u) = e^{\alpha u}$, we can write:

$$M(t)\, r_t = M(t_0)\, r_{t_0} + \alpha\beta \int_{t_0}^{t} e^{\alpha u}\, du + \sigma \int_{t_0}^{t} e^{\alpha u}\, dZ(u).$$

Divide both sides by $M(t) = e^{\alpha t}$:

$$r_t = r_{t_0} \cdot \frac{e^{\alpha t_0}}{e^{\alpha t}} + \alpha\beta \cdot \frac{1}{e^{\alpha t}} \int_{t_0}^{t} e^{\alpha u} du + \sigma \cdot \frac{1}{e^{\alpha t}} \int_{t_0}^{t} e^{\alpha u} dZ(u).$$

Simplify each term:

First term:

$$\frac{e^{\alpha t_0}}{e^{\alpha t}} = e^{-\alpha(t-t_0)} \Rightarrow r_{t_0} \cdot e^{-\alpha(t-t_0)}.$$

Second term:

$$\frac{1}{e^{\alpha t}}\int_{t_0}^{t} e^{\alpha u}du = \int_{t_0}^{t} e^{-\alpha(t-u)}du$$

Integrate the exponential function:

$$\int_{t_0}^{t} e^{\alpha u}\,du = \left[\frac{1}{\alpha}e^{\alpha u}\right]_{t_0}^{t} = \frac{1}{\alpha}\left(e^{\alpha t} - e^{\alpha t_0}\right).$$

Multiply back the constant $e^{-\alpha t}$:

$$e^{-\alpha t}\cdot\frac{1}{\alpha}\left(e^{\alpha t} - e^{\alpha t_0}\right) = \frac{1}{\alpha}\left(1 - e^{-\alpha(t-t_0)}\right).$$

$$\int_{t_0}^{t} e^{-\alpha(t-u)}\,du = \frac{1 - e^{-\alpha(t-t_0)}}{\alpha}.$$

So the second term becomes:

$$\alpha\beta\cdot\frac{1 - e^{-\alpha(t-t_0)}}{\alpha} = \beta(1 - e^{-\alpha(t-t_0)}).$$

Third term:

$$\frac{1}{e^{\alpha t}}\int_{t_0}^{t} e^{\alpha u}dZ(u) = \int_{t_0}^{t} e^{-\alpha(t-u)}dZ(u).$$

Final expression:

$$r_t = r_{t_0}e^{-\alpha(t-t_0)} + \beta(1 - e^{-\alpha(t-t_0)}) + \sigma\int_{t_0}^{t} e^{-\alpha(t-u)}dZ(u).$$

If we define $t_0 = 0$, this simplifies to:

$$r_t = (r_0 - \beta)e^{-\alpha t} + \beta + \sigma\int_{0}^{t} e^{-\alpha(t-u)}dZ(u).$$

This is the closed-form solution of the Ornstein–Uhlenbeck process.

where α is the speed of mean reversion, β the mean level attracting the interest rate (taxes) and σ is the constant volatility of interest rate. The investor takes these prices as given and chooses quantities without and with transaction costs. Further assumptions are that the securities pay dividend and taxes on the amount invested in the risky asset. Throughout this research work, we assume a probability space $(\Omega, \mathcal{F}, \mathrm{P})$ and a filtration$\{\mathcal{F}_t\}$. Uncertainty in the models is generated by standard Brownian motion$Z(t)$. The two equations governing the dynamics of the risk free and risky assets are given as;

$$dP_O(t) = (\alpha + \sigma t)P_O(t)dt \text{ or } P_O(t) = P_O(0)exp\left\{\alpha t + \frac{\sigma t^2}{2}\right\}, \tag{1}$$

and

$$dP_1(t) = P_1(t)\{\mu dt + \sigma dZ(t)\} or P_1(t) = P_1(0)exp\left\{\sigma Z(t) + \left(\mu - \frac{\sigma^2}{2}\right)t,\right\}, \forall t \in [0, \infty). \tag{2}$$

The investor is allowed to invest in the risky asset and the risk-free asset. Let $S(t)$ be the money amount invested in the risky asset at time t, then$[W(t) - S(t)]$is the money amount invested in the risk-free asset, where $W\ (t)$ is the total investment.

Clarifying Sentence Two

The phrase "the investor takes these prices as given" refers to the prices of the financial assets under consideration. Specifically, this includes the price process for the risk-free asset, denoted $P_0(t)$, which evolves according to the deterministic differential equation

$$dP_0(t) = (\alpha + \sigma t)P_0(t)\,dt,$$

with the closed-form solution

$$P_0(t) = P_0(0)\exp\left(\alpha t + \frac{\sigma t^2}{2}\right),$$

as shown in Equation (1). In addition, it refers to the stochastic price process of the risky asset, $P_1(t)$, governed by the stochastic differential equation

$$dP_1(t) = P_1(t)\left(\mu\,dt + \sigma\,dZ(t)\right),$$

with the explicit solution

$$P_1(t) = P_1(0)\exp\left(\sigma Z(t) + \left(\mu - \frac{\sigma^2}{2}\right)t\right).$$

Therefore, when the model states that "the investor takes these prices as given," it means that the investor accepts the dynamics of $P_0(t)$ and $P_1(t)$ as exogenously determined by the market. The investor does not influence or set these prices but instead chooses the quantities to invest in each asset based on their respective price evolutions. These decisions may be made either in the presence or absence of transaction costs, depending on the specific modeling assumptions.

Probability Space and Filtration

In this model, we assume a probabilistic framework formalized by the triple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω denotes the sample space consisting of all possible outcomes (for example, Ω may represent all possible paths that Brownian motion $Z(t)$ might take), $\mathcal{F}$ is a sigma-algebra on Ω representing the collection of measurable events, and $\mathbb{P}$ is a probability measure that assigns probabilities to the events in $\mathcal{F}$. This structure constitutes a probability space and provides the foundational environment in which all random processes in the model are defined.

We further assume the existence of a filtration $\{\mathcal{F}_t\}_{t\geq 0}$, which is an increasing family of sub-sigma-algebras of $\mathcal{F}$. Each $\mathcal{F}_t$ represents the information available up to and including time t. Formally, this means:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \text{for all } 0 \leq s \leq t.$$

The filtration encodes the temporal evolution of observable information and is essential for rigorously defining stochastic processes such as Brownian motion, as well as for formulating stochastic differential equations and Itô integrals.

Risk-Free Asset — $P_0(t)$

We are given the following stochastic differential equation (SDE) for the risk-free asset:

$$dP_0(t) = (\alpha + \sigma t)P_0(t)\,dt$$

The term $dP_0(t)$ represents the infinitesimal change in the value of the risk-free asset at time t. The function $P_0(t)$ denotes the current value of the asset at time t. The expression $\alpha + \sigma t$ is the instantaneous interest rate or yield, which is not constant — it starts at α and increases linearly with time at rate σ.

This means that the longer you wait, the faster the risk-free asset compounds, since the rate increases over time. The interest rate accelerates as time passes.

To put this in perspective: this is like a bank account that pays 3% interest at first, then 4%, then 5%, and so on — increasing steadily. That is what $\alpha + \sigma t$ is modeling.

We now solve the SDE:

$$dP_0(t) = (\alpha + \sigma t)P_0(t)\,dt$$

We aim to find a closed-form expression for $P_0(t)$. First, divide both sides by $P_0(t)$:

$$\frac{dP_0(t)}{P_0(t)} = (\alpha + \sigma t)\,dt$$

This is a separable differential equation, and we integrate both sides from $t = 0$ to t:

$$\int_0^t \frac{1}{P_0(s)}\,dP_0(s) = \int_0^t (\alpha + \sigma s)\,ds$$

On the left-hand side, we apply the reverse chain rule:

$$\int_0^t \frac{1}{P_0(s)}\,dP_0(s) = \ln P_0(t) - \ln P_0(0)$$

This follows from the fact that:

$$\frac{d}{ds}\ln P_0(s) = \frac{1}{P_0(s)} \cdot \frac{dP_0(s)}{ds} \quad \Rightarrow \quad d(\ln P_0(s)) = \frac{1}{P_0(s)} dP_0(s)$$

So the integral of the left-hand side is just the change in the natural logarithm of P_0 from 0 to t.
Now compute the right-hand side:

$$\int_0^t (\alpha + \sigma s)\,ds = \int_0^t \alpha\,ds + \int_0^t \sigma s\,ds = \alpha t + \frac{\sigma t^2}{2}$$

Putting both sides together:

$$\ln P_0(t) - \ln P_0(0) = \alpha t + \frac{\sigma t^2}{2}$$

Apply logarithmic identity:

$$\ln\left(\frac{P_0(t)}{P_0(0)}\right) = \alpha t + \frac{\sigma t^2}{2}$$

Exponentiate both sides:

$$\frac{P_0(t)}{P_0(0)} = \exp\left(\alpha t + \frac{\sigma t^2}{2}\right)$$

Multiply through by $P_0(0)$ to isolate $P_0(t)$:

$$P_0(t) = P_0(0) \cdot \exp\left(\alpha t + \frac{\sigma t^2}{2}\right)$$

Risky Asset — $P_1(t)$

The stochastic differential equation (SDE) governing the price of the risky asset is given by:

$$dP_1(t) = P_1(t)\left(\mu\,dt + \sigma\,dZ(t)\right).$$

This equation characterizes a geometric Brownian motion (GBM), which is the standard model used in continuous-time finance to describe the evolution of risky asset prices such as stocks. In this model, $P_1(t)$ denotes the asset price at time t, μ is the expected rate of return per unit of time (drift), and σ is the volatility, representing the magnitude of randomness in returns. The term $Z(t)$ is a standard Brownian motion (or Wiener process), and $dZ(t)$ is an infinitesimal increment satisfying $dZ(t) \sim \mathcal{N}(0, dt)$. The dt

term represents an infinitesimal time increment.

At first glance, one might ask why the current price $P_1(t)$ appears on the right-hand side of the equation, given that $dP_1(t)$ represents the instantaneous change in price. The reason lies in the nature of the GBM model: the SDE models not the absolute change in price, but the *relative*, or *proportional*, change in price. This can be seen by dividing both sides of the equation by $P_1(t)$:

$$\frac{dP_1(t)}{P_1(t)} = \mu\, dt + \sigma\, dZ(t).$$

This version of the equation states that the *fractional* change in the asset price over time is composed of a deterministic drift $\mu\, dt$ and a stochastic shock $\sigma\, dZ(t)$. Economically, this formulation captures the realistic behavior of financial assets. For example, a 1% increase in the price of a \$100 stock corresponds to a \$1 gain, while a 1% increase in a \$1 stock yields only a \$0.01 gain. The magnitude of the price change must scale with the current price, and this scaling is captured naturally by multiplying by $P_1(t)$. Without this scaling, the same absolute shock would apply equally to all assets, which contradicts how real markets behave.

Therefore, the expression $\mu\, dt + \sigma\, dZ(t)$ captures the instantaneous percentage return, and the presence of $P_1(t)$ converts that into a dollar amount of price change. This results in exponential (rather than linear) price paths, which aligns with observed behavior in financial markets.

We begin with the stochastic differential equation:

$$dP_1(t) = \mu P_1(t)\, dt + \sigma P_1(t)\, dZ(t),$$

which models a geometric Brownian motion.

This is a linear stochastic differential equation with respect to $P_1(t)$. To solve it, we apply the method of integrating factors.

We define the integrating factor:

$$M(t) = e^{-\mu t}.$$

We now multiply both sides of the SDE by the integrating factor $M(t)$, but because $P_1(t)$ is a stochastic process, we must apply the Itô product rule.

Itô's product rule: If $X(t)$ and $Y(t)$ are Itô processes, then:

$$d(X(t)Y(t)) = X(t)\, dY(t) + Y(t)\, dX(t) + dX(t)\, dY(t).$$

Apply this rule to $M(t)P_1(t)$. Since $M(t) = e^{-\mu t}$ is deterministic, we have $dM(t)\, dP_1(t) = 0$, so:

$$d(M(t)P_1(t)) = M(t)\, dP_1(t) + P_1(t)\, dM(t).$$

From this identity, we rearrange to isolate $M(t)\, dP_1(t)$:

$$M(t)\, dP_1(t) = d(M(t)P_1(t)) - P_1(t)\, dM(t).$$

This substitution allows us to rewrite the original equation while keeping track of the stochastic structure.

$$dM(t) = \frac{d}{dt} e^{-\mu t}\, dt = -\mu e^{-\mu t}\, dt = -\mu M(t)\, dt.$$

$$d(M(t)P_1(t)) = M(t)\, dP_1(t) - \mu M(t) P_1(t)\, dt.$$

Rewriting:

$$M(t)\, dP_1(t) = d(M(t)P_1(t)) + \mu M(t) P_1(t)\, dt.$$

Recall the stochastic differential equation from line two:

$$M(t)\,dP_1(t) = \mu M(t)P_1(t)\,dt + \sigma M(t)P_1(t)\,dZ(t).$$

Substitute:

$$d(M(t)P_1(t)) + \mu M(t)P_1(t)\,dt = \mu M(t)P_1(t)\,dt + \sigma M(t)P_1(t)\,dZ(t).$$

Cancel terms:

$$d(M(t)P_1(t)) = \sigma M(t)P_1(t)\,dZ(t).$$

$$\int_{t_0}^{t} d(M(u)P_1(u)) = \int_{t_0}^{t} \sigma M(u)P_1(u)\,dZ(u).$$

Left-hand side:

$$M(t)P_1(t) - M(t_0)P_1(t_0).$$

Right-hand side:

$$\int_{t_0}^{t} \sigma M(u)P_1(u)\,dZ(u).$$

Putting both sides together:

$$M(t)P_1(t) = M(t_0)P_1(t_0) + \int_{t_0}^{t} \sigma M(u)P_1(u)\,dZ(u).$$

Recall $M(t) = e^{-\mu t}$, so:

$$e^{-\mu t}P_1(t) = e^{-\mu t_0}P_1(t_0) + \int_{t_0}^{t} \sigma e^{-\mu u}P_1(u)\,dZ(u).$$

Multiply both sides by $e^{\mu t}$:

$$P_1(t) = e^{\mu(t-t_0)}P_1(t_0) + e^{\mu t}\int_{t_0}^{t} \sigma e^{-\mu u}P_1(u)\,dZ(u).$$

This is the integral-form solution, but we proceed to derive the explicit solution using Itô's Lemma.

We are given the stochastic differential equation:

$$dP_1(t) = \mu P_1(t)\,dt + \sigma P_1(t)\,dZ(t).$$

We want to compute $d(\ln P_1(t))$ using **Itô's Lemma.**

Let X_t follow:

$$dX_t = \mu_t\,dt + \sigma_t\,dW_t,$$

and let $f(t, X_t)$ be a twice continuously differentiable function. Then Itô's Lemma states:

$$df = \left(\frac{\partial f}{\partial t} + \mu_t\frac{\partial f}{\partial x} + \frac{1}{2}\sigma_t^2\frac{\partial^2 f}{\partial x^2}\right)dt + \sigma_t\frac{\partial f}{\partial x}\,dW_t.$$

Let:

$$X_t = P_1(t), \quad f(x) = \ln x.$$

Then:

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial x} = \frac{1}{x}, \quad \frac{\partial^2 f}{\partial x^2} = -\frac{1}{x^2}.$$

Apply the lemma to $f(P_1(t)) = \ln P_1(t)$:

$$d(\ln P_1(t)) = \left(0 + \mu P_1(t) \cdot \frac{1}{P_1(t)} + \frac{1}{2}\sigma^2 P_1(t)^2 \cdot \left(-\frac{1}{P_1(t)^2}\right)\right) dt$$
$$+ \sigma P_1(t) \cdot \frac{1}{P_1(t)} dZ(t)$$

$$d(\ln P_1(t)) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dZ(t)$$

Now we integrate both sides:

$$\int_{t_0}^{t} d(\ln P_1(u)) = \int_{t_0}^{t} \left(\mu - \frac{1}{2}\sigma^2\right) du + \int_{t_0}^{t} \sigma\, dZ(u).$$

Left-hand side:

$$\ln P_1(t) - \ln P_1(t_0).$$

So:

$$\ln P_1(t) = \ln P_1(t_0) + \left(\mu - \frac{1}{2}\sigma^2\right)(t - t_0) + \sigma(Z(t) - Z(t_0)).$$

Now we exponentiate to get final solution:

$$P_1(t) = P_1(t_0) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)(t - t_0) + \sigma(Z(t) - Z(t_0))\right).$$

If we define $t_0 = 0$ and $Z(0) = 0$, the formula simplifies to:

$$P_1(t) = P_1(0) \exp\left(\sigma Z(t) + \left(\mu - \frac{1}{2}\sigma^2\right) t\right), \quad \forall t \in [0, \infty).$$

Assumptions

It is assumed that transaction cost, taxes and dividends are paid on the amount invested in the risky asset.

Therefore for any policy, S, the total wealth process of the insurance company evolves according to the stochastic differential equation;

$$dW^S(t) = S(t)\frac{dP_1(t)}{P_1(t)} + (W(t) - S(t))\frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)S(t)dt. \qquad (3)$$

Explanation of the Wealth Process SDE

Consider an insurance company that invests its wealth in two assets: a risky asset with price process $P_1(t)$, and a risk-free asset with price process $P_0(t)$. Let $W(t)$ denote the total wealth at time t, and let $S(t)$ be the amount invested in the risky asset. The remainder $W(t) - S(t)$ is invested in the risk-free asset.

It is assumed that transaction costs, taxes, and dividends apply to the amount invested in the risky asset. Specifically:

- ϑ: transaction cost rate,
- β: tax rate,
- d: dividend yield.

Under a given investment policy $S(t)$, the evolution of the total wealth $W^S(t)$ follows the stochastic differential equation:

$$dW^S(t) = S(t)\frac{dP_1(t)}{P_1(t)} + (W(t) - S(t))\,\frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)\,S(t)\,dt. \tag{1}$$

- $S(t)\frac{dP_1(t)}{P_1(t)}$: Gain (or loss) from the portion of wealth invested in the risky asset.
- $(W(t) - S(t))\,\frac{dP_0(t)}{P_0(t)}$: Gain from the portion of wealth invested in the risk-free asset.
- $-(\vartheta + \beta - d)\,S(t)\,dt$: Net costs associated with the risky investment due to transaction costs and taxes, partially offset by dividends. The expression $(\vartheta + \beta - d)S(t)$ represents the instantaneous rate of net cost incurred from holding the risky asset, due to transaction costs and taxes (offset by dividends). Multiplying this rate by dt gives the total cost incurred over the small time interval dt.

This equation models the dynamics of wealth by accounting for the returns from both assets, while subtracting the frictions introduced by real-world investment costs.

In ordinary calculus, if you know that a function $f(t)$ has a derivative $f'(t)$, then over a small time dt, the change in f is approximately:

$$df(t) = f'(t)\,dt$$

Here it's the same idea. You're saying that the change in some process is:

$$d(\text{some variable}) = \left(\frac{\delta(t)}{t} - (\vartheta + \beta - d)S(t)\right)dt$$

So dt is needed to **scale** the rate to an actual small increment in the process.

Substituting the expressions for, $\frac{dP_1(t)}{P_1(t)}$, and$\frac{dP_O(t)}{P_O(t)}$, the stochastic differential equation for the wealth process of the investor becomes;

$$dW^S(t) = \{[W(t)(\alpha + \sigma t) + [(\mu + d - (\alpha + \sigma t) - \vartheta - \beta)S(t)]\}dt + \sigma S(t)dZ(t). \tag{4}$$

Here, the rate of the taxes in the financial market is $\beta$and $d$is the dividend income. The rate of transaction costs is ϑ. Correspondingto a trading strategy $S(t)$and an initial capital,$W(0)$, the wealth process $W(t)$of the investor follows (4)

Let us now derive the stochastic differential equation for the investor's wealth process $dW^S(t)$ step by step, starting from the dynamics of the asset prices and the portfolio allocation strategy, and using the exact formulation provided in the paper.

The paper defines the total wealth process as:

$$dW^S(t) = S(t)\frac{dP_1(t)}{P_1(t)} + (W(t) - S(t))\,\frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)S(t)\,dt$$

This formulation explicitly includes the correction term for dividend income and taxation frictions on the risky asset.

From earlier in the paper, we are given the dynamics of the risky and risk-free assets as:

$$\frac{dP_1(t)}{P_1(t)} = \mu\,dt + \sigma\,dZ(t), \quad \frac{dP_0(t)}{P_0(t)} = (\alpha + \sigma t)\,dt$$

Substituting these expressions into the wealth process gives:

$$dW^S(t) = S(t)\,(\mu\,dt + \sigma\,dZ(t)) + (W(t) - S(t))\,(\alpha + \sigma t)\,dt - (\vartheta + \beta - d)S(t)\,dt$$

We now expand all terms:

$$dW^S(t) = \mu S(t)\,dt + \sigma S(t)\,dZ(t) + (\alpha + \sigma t)W(t)\,dt - (\alpha + \sigma t)S(t)\,dt - (\vartheta + \beta - d)S(t)\,dt$$

We now collect all dt terms:

$$dW^S(t) = \Big[(\alpha + \sigma t)W(t) + \mu S(t) - (\alpha + \sigma t)S(t) - (\vartheta + \beta - d)S(t)\Big]dt + \sigma S(t)\,dZ(t)$$

Grouping the terms involving $S(t)$, we obtain:

$$dW^S(t) = [(\alpha + \sigma t)W(t) + (\mu - (\alpha + \sigma t) - \vartheta - \beta + d)\,S(t)]\,dt + \sigma S(t)\,dZ(t)$$

Thus, the final fully justified result is:

$$dW^S(t) = \{W(t)(\alpha + \sigma t) + [\mu + d - (\alpha + \sigma t) - \vartheta - \beta]\,S(t)\}\,dt + \sigma S(t)\,dZ(t)$$

This expression exactly matches Equation (4) in the paper. The terms ϑ, β, and d are not artificially introduced in Step 4; they are already embedded in the wealth process definition and arise naturally through direct substitution and simplification.

Suppose the investor has a utility function$U(.)$ which is strictly concave and continuously differentiableon $(-\infty; +\infty)$andwishes to maximize the expected utility of his terminal wealth. Then, the investor's problem can therefore be written as;

$$\left.\begin{array}{r} \mathrm{Max}_S\{\mathcal{B}^S V(\mathrm{t}, w)\} = 0;\ V(T, w) = U(w) \\ V(T, w) = \underset{S}{\mathrm{Max}} E^{t,w}|U(w^s(t))| \end{array}\right\} \quad (5)$$

where $\mathcal{B}^S$ is an operator that will be obtained using Ito's lemma, subject to (4).

The $\max_S$ noation is telling us that among all possible controls $S(t)$, find the one that gives you the **best** expected infinitesimal change in value — and for the optimal $S^*(t)$, that best rate of change **must be zero**.

Interpretation

We focus exclusively on the equation:

$$\max_S \left\{\mathcal{B}^S V(t, w)\right\} = 0$$

This equation characterizes the optimal behavior of an investor in continuous time. Each term in the equation plays a specific role in the dynamic optimization framework.

The control variable S represents the investment strategy. At each time t, the investor decides how to allocate current wealth between risky and risk-free assets. For example, one might choose $S(t) = 0.6$ to invest 60% in a stock and 40% in a bond, or $S(t) = 1$ to allocate all wealth to the risky asset. In general, the set of admissible strategies may allow for leverage or short selling, i.e., $S(t) \in (-\infty, \infty)$.

The value function $V(t, w)$ represents the maximum expected utility an investor can attain, conditional on starting at time t with current wealth w, and making optimal investment decisions from that point forward.

Formally, it is defined as:

$$V(t, w) = \max_{S} \mathbb{E}^{t,w} \left[U(w^S(T)) \right]$$

We now define and interpret each term appearing in this equation:

- $t \in [0, T]$ denotes the current time, where T is the terminal time horizon (e.g., retirement or death). This is the point in time from which the investor begins optimizing their investment decisions.
- $w \in \mathbb{R}_+$ denotes the investor's current wealth at time t.
- $S = \{S_u\}_{u \in [t,T]}$ is a control process representing the investor's strategy over time. Specifically, S_u denotes the proportion of wealth invested in the risky asset at time u. The control is adapted to the available information and can vary with time and market conditions.
- The quantity $w^S(T)$ denotes the investor's terminal wealth at time T, conditional on following strategy S starting from time t. It is a random variable because the evolution of the wealth process $W^S(t)$ depends on the stochastic dynamics of the risky and risk-free asset prices, denoted $P_1(t)$ and $P_0(t)$, respectively.

$$V(t, w) = \max_{S} \int_{\text{all paths}} U\big(w^S(T, \omega)\big) \, \mathbb{P}^{t,w}(d\omega)$$

- $V(t, w)$ — Value function: maximum expected utility of terminal wealth starting at time t with wealth w.
- $\max_S$ — Optimization over all admissible control strategies S (portfolio allocations).
- $\int_{\text{all paths}} \cdots \mathbb{P}^{t,w}(d\omega)$ — Integration over all possible realizations ω of the stochastic process, where each outcome is weighted by the probability of that path occurring.
- $U(\cdot)$ — Utility function mapping terminal wealth to a measure of preference or satisfaction.
- $w^S(T, \omega)$ — Terminal wealth at time T given strategy S and realized path ω (this is the "value" of that path).

$$w^S(T, \omega) = w \ + \int_t^T \Big\{ W(u)(\alpha + \sigma u) + [(\mu + d) - (\alpha + \sigma u) - \vartheta - \beta] \, S(u) \Big\} \, du \ + \int_t^T \sigma \, S(u) \, dZ_u(\omega)$$

- $\mathbb{P}^{t,w}(d\omega)$ — Probability that the specific path ω occurs, given starting time t and wealth w (this is the "probability" of that path).
- The paper defines the wealth process as:

$$dW^S(t) = S(t) \frac{dP_1(t)}{P_1(t)} + (W(t) - S(t)) \, \frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d) \, S(t) \, dt$$

- $S(t)$ — Amount invested in the risky asset at time t.
- $W(t) - S(t)$ — Amount invested in the risk-free asset at time t.
- $\frac{dP_1(t)}{P_1(t)}$ — Instantaneous return of the risky asset.
- $\frac{dP_0(t)}{P_0(t)}$ — Instantaneous return of the risk-free asset.
- The price processes $P_1(t)$ and $P_0(t)$ are stochastic (often modeled with Brownian motion), so portfolio returns—and total wealth—are random over time.

- Even if the control S is deterministic, the evolution of $W^S(t)$ and terminal wealth $w^S(T)$ is random because of asset price uncertainty.
- This randomness propagates through the wealth process and affects the realized utility $U(w^S(T))$, which is evaluated in expectation in the value function.
- $U(\cdot)$ is the investor's utility function, which maps terminal wealth to a real number reflecting the investor's preferences. It is assumed to be increasing (more wealth is preferred) and concave (reflecting risk aversion). Examples include:
$$U(w) = \log(w) \quad \text{or} \quad U(w) = \frac{w^{1-\gamma}}{1-\gamma}, \quad \gamma \neq 1.$$
- $\mathbb{E}^{t,w}[\cdot]$ denotes conditional expectation, taken over all sources of randomness in the market, given that the current time is t and the investor's wealth is w. It reflects the investor's forecast of future utility based on known information at time t.
- $\max_S$ indicates that the investor considers all admissible strategies S, and chooses the one that maximizes the conditional expected utility of terminal wealth.

In summary, the value function $V(t,w)$ captures the best possible expected utility of final wealth that the investor can hope to achieve, starting from the current state (t,w) and acting optimally through time T.

The operator $\mathcal{B}^S$ arises from applying Itô's lemma to the value function. It captures the infinitesimal expected change in $V(t,w)$ under the dynamics of the wealth process and depends on the strategy S. It typically has the following form:
$$\mathcal{B}^S V(t,w) = \frac{\partial V}{\partial t} + \mu^S(t,w)\frac{\partial V}{\partial w} + \frac{1}{2}\left(\sigma^S(t,w)\right)^2 \frac{\partial^2 V}{\partial w^2}$$
Here, μ^S denotes the drift of the wealth process under strategy S, and σ^S is the associated volatility. The first derivative $\frac{\partial V}{\partial w}$ captures the marginal value of wealth, while the second derivative $\frac{\partial^2 V}{\partial w^2}$ reflects the curvature (risk sensitivity) of the value function.

The maximization $\max_S \left\{\mathcal{B}^S V(t,w)\right\}$ selects the strategy S that maximizes the expected infinitesimal improvement in the value function. Since the value function is defined as the maximum over all such strategies, the optimal choice must satisfy the condition that this maximum is equal to zero. If $\mathcal{B}^S V(t,w) < 0$, the investor is not maximizing utility; if it is positive, the strategy can be improved. Thus, at the optimal strategy, the condition $\max_S \left\{\mathcal{B}^S V(t,w)\right\} = 0$ must hold.

Intuitively, the investor stands on a landscape (the value surface) and chooses the direction (strategy) that locally increases expected utility the most. At the optimal point, no further improvement is possible; the slope is zero in the best direction.

Step-by-Step View

- **Start the process at terminal time T**:
You know:
$$V(T,w) = U(w)$$
because *no further decisions* are made beyond this point—utility is realized.
- **Step back a small time**:
Let $t = T - \Delta t$, and suppose your wealth at this earlier time is w.
- **Forward simulate from t to T** under a *candidate strategy S*:
That gives:
$$W^S(T) = w + \int_t^T dW^S(u)$$
This is your **random terminal wealth** under strategy S, starting from wealth w.

- **Plug terminal wealth into utility function**:

$$U(W^S(T))$$

is a *random variable* representing the realized utility if you follow strategy S from now until T.

- **Take expectation** to account for randomness in asset prices:

$$\mathbb{E}^{t,w}\left[U(W^S(T))\right]$$

gives you the **expected utility** from following strategy S starting at wealth w.

- Compute the infinitesimal generator operator $\mathcal{B}^S V(t,w)$, which gives the **expected** *infinitesimal rate of change* in value under strategy S.

  Solve:

$$\max_S \mathcal{B}^S V(t,w) = 0$$

Note: You're *not* trying to get a numerical value of zero; rather, you're finding the strategy S^* that makes this equation hold. The zero indicates that no local improvement is possible under the optimal policy. The strategy can be found using *equation (12)*.

- **You've now determined the optimal strategy to follow from time t to T** given that you currently have wealth w.

- **Step back again**:
  Let $t = T - 2\Delta t$, and repeat this process recursively.

Solving the Infinitesimal Generator in Practice

Solving the infinitesimal generator involves a large number of calculations. In theory, it is an infinite-dimensional problem because time, wealth, and control are continuous variables. In practice, the problem is made tractable by discretizing everything (time, wealth, and strategies).
What the Computer Actually Does

- Build a 2D table of size $N \times M$ (time $\times$ wealth).

- Start at the last time step. Set the terminal condition:

$$V(T, w_i) = U(w_i)$$

- Step backward through time. At each point (t_k, w_i), solve:

$$\max_S \mathcal{B}^S V(t_k, w_i)$$

- Store the resulting value $V(t_k, w_i)$.

- Store the optimal control $S^*(t_k, w_i)$.

Computational Cost

- At each grid point (t_k, w_i), evaluate a set of possible strategies $S_1, S_2, \ldots, S_L$.

- For each strategy S_ℓ, compute:

$$\mathcal{B}^{S_\ell} V(t_k, w_i)$$

- Take the maximum over all of them:

$$V(t_k, w_i) = \max_\ell \mathcal{B}^{S_\ell} V(t_k, w_i)$$

- The total number of operations scales like:

$$\text{Cost} \sim (\#\text{ strategies}) \times (\#\text{ wealth points}) \times (\#\text{ time steps})$$

Example

- 500 wealth grid points
- 100 time steps
- 50 candidate strategies per point
- Total operations: $500 \times 100 \times 50 = 2.5$ million evaluations

Final Intuition

- You do not track wealth forward through time.
- You compute how the value function behaves at each possible wealth level and time.
- The more grid points w_i you use, the more accurate the solution becomes.
- More grid points means more compute.
- You're building a table of optimal actions for every case.

The quadratic variation of the wealth process is;

$$< dW^S(t)) >= S^2(t)\sigma^2 dt. \quad (6)$$

The optimization problem being considered is for the power utility function given in the form;

$$U(w) = \frac{w^{1-\phi} - k}{1-\phi}, \quad (7)$$

where ϕ and k are constants and, $\phi \neq 1$.

Quadratic Variation

The expression

$$\langle dW^S(t) \rangle = S^2(t)\sigma^2 dt$$

does **not** represent the classical variance of a random variable. Instead, it is shorthand from stochastic calculus that describes the **quadratic variation** of the wealth process $W^S(t)$.

In stochastic calculus, the quadratic variation $\langle W^S \rangle_t$ of a process $W^S(t)$ tracks how much "randomness" accumulates over time. It replaces the classical notion of variance when working with continuous-time processes.

You're saying:

- "The process $W^S(t)$ accumulates variance at a rate of $S^2(t)\sigma^2$ per unit time."

It's like saying:

$$\frac{d}{dt} \langle W^S \rangle_t = S^2(t)\sigma^2$$

and you're just expressing this "per infinitesimal unit of time" using differential notation.

Note: The notation $\langle W^S \rangle_t$ refers to the quadratic variation of the process $W^S(t)$ *accumulated up to time* t. It represents the total variance that has built up over the interval $[0, t]$.

Interpretation: Taking the derivative $\frac{d}{dt}\langle W^S \rangle_t$ tells us the *instantaneous rate* at which the quadratic variation — i.e., accumulated variance — is growing at time t. In this case, the rate is given by $S^2(t)\sigma^2$.

Optimal Investment Strategy for the Power Utility Function

In this section, the explicit solutions for the optimization problem are obtained using stochastic control and Ito lemma.

General Framework

Define the value function as;

$$J(t, s, w) = \mathrm{Max}_S\{\mathcal{B}^S V(t, w)\} = 0;\ V(T, w) = U(w), 0 < t < T. \tag{8}$$

The corresponding HJB equation is given by;

$$J_t + J_w\{[W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t + \vartheta + \beta)]S(t)]\} + J_{ww}\left[\frac{S^2(t)\sigma^2}{2}\right] = 0, \tag{9}$$

where J_t, J_w, and J_{ww} denote the first partial derivatives of J respect to t, the wealth w respectively and second order partial derivatives with respect to the wealth w, with the boundary condition that at the terminal time T,

x

Deriving the HJB Equation

We consider the standard scalar version of **Itô's Lemma** for a function $f(t, X_t)$, where the process X_t satisfies:

$$dX_t = \mu_t\, dt + \sigma_t\, dW_t$$

Then, for a twice continuously differentiable function $f(t, X_t)$, Itô's Lemma gives:

$$df = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2}\sigma_t^2 \frac{\partial^2 f}{\partial x^2}\right) dt + \sigma_t \frac{\partial f}{\partial x}\, dW_t$$

Let:

- $X_t = W^S(t)$: the wealth process,
- $f(t, X_t) = V(t, W(t))$: the value function,
- μ_t and σ_t: the drift and diffusion from the wealth SDE (Equation (4)).

From the paper, the wealth process satisfies:

$$dW(t) = \underbrace{[W(t)(\alpha + \sigma t) + ((\mu + d) - (\alpha + \sigma t) - \vartheta - \beta)S(t)]}_{\mu_t}\, dt + \underbrace{\sigma S(t)}_{\sigma_t}\, dZ(t)$$

Using:

$$dV = \left(\frac{\partial V}{\partial t} + \mu_t \frac{\partial V}{\partial w} + \frac{1}{2}\sigma_t^2 \frac{\partial^2 V}{\partial w^2} \right) dt + \sigma_t \frac{\partial V}{\partial w} dZ_t$$

Substitute:

$$\mu_t = W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t) - \vartheta - \beta]\, S(t)$$
$$\sigma_t = \sigma S(t)$$

We get:

$$\begin{aligned} dV(t, W(t)) = \bigg[&\frac{\partial V}{\partial t} + \frac{\partial V}{\partial w} \cdot (W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t) - \vartheta - \beta]\, S(t)) \\ &+ \frac{1}{2} \cdot \frac{\partial^2 V}{\partial w^2} \cdot \sigma^2 S^2(t) \bigg] dt + \sigma S(t) \cdot \frac{\partial V}{\partial w} \cdot dZ(t) \end{aligned}$$

We have a differential expression involving a stochastic term:

$$\frac{\partial V}{\partial w} \cdot \sigma S(t)\, dZ(t)$$

This term is a martingale increment, and since $dZ(t) \sim \mathcal{N}(0, dt)$, it is normally distributed with mean centered at 0. Therefore:

$$\mathbb{E}\left[\frac{\partial V}{\partial w} \cdot \sigma S(t)\, dZ(t) \right] = \mathbb{E}\left[\frac{\partial V}{\partial w} \cdot \sigma S(t) \right] \cdot \mathbb{E}[dZ(t)] = 0$$

The remaining terms are multiplied by dt, and since they are constant (deterministic), their expectation is just the term itself. So when we take the expectation of the full differential dV, we get:

$$\mathbb{E}[dV] = \left(\frac{\partial V}{\partial t} + \frac{\partial V}{\partial w} \cdot (W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t) - \vartheta - \beta]\, S(t)) + \frac{1}{2} \cdot \frac{\partial^2 V}{\partial w^2} \cdot \sigma^2 S^2(t) \right) dt$$

We observe that since the expectation of the change in value $\mathbb{E}[dV]$ must be zero (by the Bellman principle), the expression multiplied by dt must itself be zero. That gives the HJB:

$$\max_S \left\{ \frac{\partial V}{\partial t} + \frac{\partial V}{\partial w} \cdot (W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t) - \vartheta - \beta]\, S(t)) + \frac{1}{2} \cdot \frac{\partial^2 V}{\partial w^2} \cdot \sigma^2 S^2(t) \right\} = 0$$

Why You Must Eliminate the Noise Term in the HJB Derivation

If we do *not* eliminate the random noise term in the differential of the value function,

$$dV = \left(V_t + V_w \mu + \frac{1}{2} V_{ww} \sigma^2 \right) dt + V_w \sigma\, dZ_t,$$

then we are trying to optimize a quantity that is **not deterministic**.

- The term dZ_t represents Brownian motion and varies randomly with each realization.
- As a result, $dV(t, W(t))$ is a stochastic quantity — it is not fixed, and changes every time the noise path changes.
- Attempting to optimize this expression leads to solutions that depend on random noise rather than on the control $S(t)$.
- In other words, you would get a different "optimal" control each time, purely due to randomness — which defeats the entire purpose of solving for optimality.

Taking expectation eliminates the stochastic term:

$$\mathbb{E}[V_w \sigma \, dZ_t] = 0,$$

leaving only the deterministic drift:

$$\mathbb{E}[dV] = \left(V_t + V_w \mu + \frac{1}{2} V_{ww} \sigma^2 \right) dt.$$

This expression is now deterministic and suitable for optimization. It reflects how the expected value of the objective function evolves over time under a given strategy.

Bellman's principle says that if you are already behaving optimally, then you cannot expect to gain any additional value by waiting. That means the **expected infinitesimal change** in the value function must vanish.

$$J(T, s, w) = U(w). \tag{10}$$

The differentiation of (9) with respect to $S(t)$ gives the optimal policy;

$$J_w(\mu + d - (\alpha + \sigma t) - \vartheta - \beta) + \sigma^2 S(t) J_{ww} = 0. \tag{11}$$

This simplifies to,

$$S^*(t) = \frac{[(\vartheta + \beta + \alpha + \sigma t) - (\mu + d)] J_w}{\sigma^2 J_{ww}}. \tag{12}$$

Derivation of the Optimal Control $S(t)$

We're optimizing an expression **pointwise in time** — meaning at each fixed (t, w), we treat the function:

$$F(S) = J_w \cdot [\mu + d - (\alpha + \sigma t) - \vartheta - \beta] + \frac{1}{2} J_{ww} \cdot \sigma^2 S^2$$

as a function of only $S(t)$, holding t, w, J_w, and J_{ww} fixed. At that single point in time, this becomes a **real-valued function of one real variable**, so you **can** take a derivative and apply the standard first-order optimality condition:

$$\frac{dF}{dS} = 0$$

This is *exactly like* a single-variable optimization problem **at each point in the state space**.

Variable Definitions (from the paper)

- μ: mean return on the risky asset
- d: **dividend income**
- $\alpha + \sigma t$: **return on the risk-free asset**, where α is the base rate and σ is the slope of the interest rate
- ϑ: **transaction cost rate** (proportional to risky asset investment)
- β: **tax rate** (financial market tax, also proportional to risky asset investment)
- $S(t)$: amount invested in the risky asset at time t
- J_w: partial derivative of the value function with respect to wealth
- J_{ww}: second partial derivative of the value function with respect to wealth
- σ: volatility of the risky asset

The Optimal Strategy for the Power Utility Function

Considering the power utility function described by (7), to eliminate dependency on $w$let the solution to the HJB equation (9) is of the form,

$$J(t,s,w) = h(t,T)\frac{w^{1-\phi}-k}{1-\phi}, \tag{13}$$

with the boundary condition,

$$h(T,T) = 1. \tag{14}$$

Then,

$$J_t = h_t\frac{w^{1-\phi}-k}{1-\phi}, J_w = hw^{-\emptyset}, \text{ and } J_{ww} = -\emptyset w^{-\emptyset-1}. \tag{15}$$

The HJB equation (9) becomes;

Eliminating the Dependency on w

"To eliminate dependency on w, let the solution to the HJB equation (9) be of the form..."

This phrase refers to a simplification technique used in solving the Hamilton-Jacobi-Bellman (HJB) equation:

$$J_t + J_w \cdot [W(t)(\alpha + \sigma t) + ((\mu + d) - (\alpha + \sigma t) - \vartheta - \beta)\, S(t)] + J_{ww} \cdot \frac{S^2(t)\sigma^2}{2} = 0 \tag{9}$$

This equation depends on both t and w, making it a partial differential equation (PDE) that is generally difficult to solve directly.

At terminal time T, the value function is known:

$$J(T,s,w) = V(T,w) = U(w) = \frac{w^{1-\phi}-k}{1-\phi} \tag{7}$$

This provides a boundary condition and reveals the functional form of J at $t = T$.

At terminal time T, the process ends. There is no further control to optimize. As a result, the value function is no longer governed by the infinitesimal expected change in $V(t,w)$; it simply equals the realized utility of the terminal wealth:

$$J(T,s,w) = U(w)$$

This serves as the boundary condition for solving the Hamilton–Jacobi–Bellman (HJB) equation backward in time.

Because $J(T,s,w)$ is proportional to $w^{1-\phi}$, it is natural to assume that the entire function $J(t,s,w)$ maintains the same structure with respect to w for all earlier times $t < T$, scaled by some unknown function of time. This motivates the following ansatz:

$$J(t,s,w) = h(t,T) \cdot \frac{w^{1-\phi}-k}{1-\phi} \tag{13}$$

Ansatz: an assumption about the form of an unknown function which is made in order to facilitate solution of an equation or other problem.

By proposing the form in (13), the authors aim to factor out the w-dependence from the HJB equation. To do this, they compute the partial derivatives:

$$\begin{aligned} J_t &= h_t \cdot \frac{w^{1-\phi} - k}{1-\phi} \\ J_w &= h(t,T) \cdot w^{-\phi} \\ J_{ww} &= -\phi \cdot h(t,T) \cdot w^{-\phi-1} \end{aligned} \tag{15}$$

Each term is now a product of a function of t and a power of w. When these expressions are substituted back into Equation (9), every term will include a factor of $w^{1-\phi}$, which can be canceled out.

As a result, the entire PDE simplifies into an ordinary differential equation (ODE) in t for the unknown function $h(t,T)$, with all dependence on w eliminated.

$$h_t \frac{w^{1-\phi}-k}{1-\phi} + hw^{-\emptyset}\{[W(t)(\alpha+\sigma t) + [(\mu+d) - (\alpha+\sigma t+\vartheta+\beta)]S^*(t)]\} + \sigma S^*(t)dZ(t) - \emptyset w^{-\emptyset-1} h \left[\frac{S^{*2}(t)\sigma^2}{2}\right] = 0, \tag{16}$$

which simplifies to,

$$\frac{h_t}{h} = -\left\{\begin{matrix} \frac{(1-\phi)w^{-\emptyset}}{w^{1-\phi}-k}\{[W(t)(\alpha+\sigma t) + [(\mu+d) - (\alpha+\sigma t+\vartheta+\beta)]S^*(t)]\} - \\ \frac{(1-\phi)\emptyset w^{-\emptyset-1}}{w^{1-\phi}-k}\left[\frac{S^{*2}(t)\sigma^2}{2}\right] \end{matrix}\right\}. \tag{17}$$

Deriving Equation (17)

We start from equation (16):

$$\frac{h_t w^{1-\phi} - k}{1-\phi} + hw^{-\phi}\left\{W(t)(\alpha+\sigma t) + [(\mu+d) - (\alpha+\sigma t+\vartheta+\beta)]\, S^*(t)\right\} + \sigma S^*(t)dZ(t) - \phi h w^{-\phi-1} \cdot \frac{S^*(t)^2\sigma^2}{2} = 0 \tag{16}$$

We define the components separately:

$$A(t) := W(t)(\alpha+\sigma t), \quad B(t) := (\mu + d - \alpha - \sigma t - \vartheta - \beta)$$

Then equation (16) becomes:

$$\frac{h_t w^{1-\phi} - k}{1-\phi} + hw^{-\phi}\left(A(t) + B(t)S^*(t)\right) + \sigma S^*(t)dZ(t) - \phi h w^{-\phi-1} \cdot \frac{S^*(t)^2\sigma^2}{2} = 0$$

Taking expectations (i.e., dropping the martingale term involving $dZ(t)$), and dividing both sides by h:

$$\frac{h_t}{h} \cdot \frac{w^{1-\phi} - k}{1-\phi} + w^{-\phi}A(t) + w^{-\phi}B(t)S^*(t) - \phi w^{-\phi-1} \cdot \frac{S^*(t)^2\sigma^2}{2} = 0$$

This is the simplified and correctly factored version:

$$\frac{h_t}{h} = -\left(\frac{(1-\phi)w^{-\phi}}{w^{1-\phi}-k}A(t) + \frac{(1-\phi)w^{-\phi}}{w^{1-\phi}-k}B(t)S^*(t) - \frac{(1-\phi)\phi w^{-\phi-1}}{w^{1-\phi}-k} \cdot \frac{S^*(t)^2\sigma^2}{2}\right) \tag{17}$$

Differentiating (16) with respect to $S^*(t)$ obtained the optimal investment in the risky asset as;

$$S_{d,\vartheta,\beta}{}^*(t) = \frac{[(\mu+d)-(\alpha+\sigma t+\vartheta+\beta)]}{\emptyset\sigma^2} w \tag{18}$$

Deriving Equation (18)

We start from the expected version of the HJB Equation (16), with the stochastic term removed:

$$\frac{h_t w^{1-\phi}}{1-\phi} - k + hw^{-\phi}\left\{W(t)(\alpha+\sigma t) + [(\mu+d) - (\alpha+\sigma t+\vartheta+\beta)]\,S^*(t)\right\} - \phi h w^{-\phi-1}\cdot\frac{S^*(t)^2\sigma^2}{2} = 0 \quad (16)$$

Let us define:

$$A(t) := W(t)(\alpha+\sigma t), \quad B(t) := (\mu+d) - (\alpha+\sigma t+\vartheta+\beta)$$

Substituting into the equation:

$$\frac{h_t w^{1-\phi}}{1-\phi} - k + hw^{-\phi}\left(A(t) + B(t)S^*(t)\right) - \phi h w^{-\phi-1}\cdot\frac{S^*(t)^2\sigma^2}{2} = 0$$

We now take the derivative with respect to the control variable $S^*(t)$:

$$\frac{d}{dS^*(t)}\left[hw^{-\phi}B(t)S^*(t) - \phi h w^{-\phi-1}\cdot\frac{S^*(t)^2\sigma^2}{2}\right] = 0$$

Differentiating term by term:

$$hw^{-\phi}B(t) - \phi h w^{-\phi-1}\cdot\sigma^2 S^*(t) = 0$$

Cancel h from both terms:

$$w^{-\phi}B(t) = \phi w^{-\phi-1}\cdot\sigma^2 S^*(t)$$

Multiply both sides by w:

$$w^{1-\phi}B(t) = \phi w^{-\phi}\cdot\sigma^2 S^*(t)$$

Divide both sides by $\phi w^{-\phi}\sigma^2$:

$$S^*(t) = \frac{B(t)}{\phi\sigma^2}\cdot w = \frac{(\mu+d) - (\alpha+\sigma t+\vartheta+\beta)}{\phi\sigma^2}\cdot w$$

Final Result:

$$S^*_{d,\vartheta,\beta}(t) = \frac{(\mu+d) - (\alpha+\sigma t+\vartheta+\beta)}{\phi\sigma^2}\cdot w$$

From equation (17) obtained;

$$\int_t^T \frac{dh}{h} ds = \int_t^T (A+Bs)ds, \quad (19a)$$

Where

$$(A+Bt) = -\left\{\frac{(1-\phi)w^{-\emptyset}}{w^{1-\phi}-k}\{[W(t)(\alpha+\sigma t) + [(\mu+d) - (\alpha+\sigma t+\vartheta+\beta)]S^*(t)]\} - \frac{(1-\phi)\emptyset w^{-\emptyset-1}}{w^{1-\phi}-k}\left[\frac{S^{*2}(t)\sigma^2}{2}\right]\right\}. \quad (19b)$$

The solution of equation (19a) gives;

$$\ln\left[\frac{h(T,T)}{h(t,T)}\right] = A(T-t) + \frac{B(T^2-t^2)}{2}, \quad (20)$$

which simplifies to

$$h(t,T) = h(T,T)\exp\left\{-[A(T-t) + \frac{B(T^2-t^2)}{2}]\right\}. \quad (21a)$$

Clean, Corrected, and Step-by-Step Derivation of Equation (20)

We begin with *equation (17)*:

$$\frac{h_t}{h} = -\left(\frac{(1-\phi)w^{-\phi}}{w^{1-\phi}-k} W(t)(\alpha+\sigma t) + \frac{(1-\phi)w^{-\phi}}{w^{1-\phi}-k} \left[(\mu+d)-(\alpha+\sigma t+\vartheta+\beta)\right] S^*(t) - \frac{(1-\phi)\phi w^{-\phi-1}}{w^{1-\phi}-k} \cdot \frac{S^*(t)^2\sigma^2}{2} \right)$$

From the first-order condition, we have:

$$S^*(t) = \frac{[(\mu+d)-(\alpha+\sigma t+\vartheta+\beta)]w}{\phi\sigma^2}$$

Let's define constant factors to simplify notation:

$$C_1 := \frac{(1-\phi)w^{-\phi}}{w^{1-\phi}-k}$$
$$C_2 := \frac{(1-\phi)\phi w^{-\phi-1}}{w^{1-\phi}-k}$$

Then *equation (17)* becomes:

$$\frac{h_t}{h} = -\left[C_1 W(t)(\alpha+\sigma t) + C_1 \left[(\mu+d)-(\alpha+\sigma t+\vartheta+\beta)\right] S^*(t) - C_2 \cdot \frac{S^*(t)^2\sigma^2}{2} \right]$$

To match the structure in the paper:

$$A(t) := C_1 W(t)(\alpha+\sigma t)$$
$$B(t) := C_1 \left[(\mu+d)-(\alpha+\sigma t+\vartheta+\beta)\right] S^*(t) - C_2 \cdot \frac{S^*(t)^2\sigma^2}{2}$$

Then Equation (17') becomes:

$$\frac{h_t}{h} = -A(t) - B(t)$$

Integrate the differential equation:

$$\int_t^T \frac{h'_s}{h(s)} ds = -\int_t^T (A(s)+B(s))\, ds$$

Left-hand side simplifies:

$$\ln\left(\frac{h(T)}{h(t)}\right) = -\int_t^T A(s)ds - \int_t^T B(s)ds$$

Thus:

$$\ln\left(\frac{h(T)}{h(t)}\right) = -\int_t^T \left[A(s)+B(s)\right] ds$$

Fully Substituted Version:

$$\begin{aligned}\ln\left(\frac{h(T)}{h(t)}\right) = -\int_t^T \Bigg[& \frac{(1-\phi)w^{-\phi}}{w^{1-\phi}-k} W(s)(\alpha+\sigma s) \\ & + \frac{(1-\phi)w^{-\phi}}{w^{1-\phi}-k} \left[(\mu+d)-(\alpha+\sigma s+\vartheta+\beta)\right] \cdot \frac{[(\mu+d)-(\alpha+\sigma s+\vartheta+\beta)]w}{\phi\sigma^2} \\ & - \frac{(1-\phi)\phi w^{-\phi-1}}{w^{1-\phi}-k} \cdot \frac{1}{2} \cdot \left(\frac{[(\mu+d)-(\alpha+\sigma s+\vartheta+\beta)]w}{\phi\sigma^2} \right)^2 \sigma^2 \Bigg] ds\end{aligned}$$

- $A(s)$ and $B(s)$ are both **functions of time** through $W(s)$, σs, and $S^*(s)$.
- If, under simplifying assumptions, the paper models $A(s)$ and $B(s)$ as **linear in time**, say:

$$A(s) + B(s) = a_0 + a_1 s$$

then:

$$\int_t^T (a_0 + a_1 s)ds = a_0(T - t) + \frac{a_1}{2}(T^2 - t^2)$$

which explains the presence of $T - t$ and $T^2 - t^2$ in equation (21a).

Linear Approximation Accuracy for Small Time Intervals

When the integral interval $[t, T]$ becomes very small (i.e., $T - t \to 0$), the **linear approximation becomes much more accurate**.

Taylor Expansion: Any smooth function $f(s)$ can be expanded around point t:

$$f(s) = f(t) + f'(t)(s - t) + \frac{f''(t)}{2}(s - t)^2 + O((s - t)^3)$$

As $T - t \to 0$: The quadratic and higher-order terms become **negligible** compared to the linear term.

Practical Implications:

This suggests the paper's approach might be:

- **Short-term approximation**: Valid for small time horizons
- **Continuous-time limit**: As $dt \to 0$, the linear approximation becomes exact
- **Discrete-time implementation**: Use small time steps where linearization is accurate

Therefore, the smaller the integration interval, the better the linear approximation works!

Note: So $B(s)$ represents the time-dependent (linear) part of the model, while $A(s)$ would be the more constant-like part. But together they're approximated as $a_0 + a_1 s$.

Simplification Steps

We start from equation (20):

$$\ln\left[\frac{h(T,T)}{h(t,T)}\right] = A(T - t) + \frac{B(T^2 - t^2)}{2}$$

Using the logarithmic identity

$$\ln\left[\frac{h(T,T)}{h(t,T)}\right] = -\ln\left[\frac{h(t,T)}{h(T,T)}\right],$$

we can rewrite equation (20) as

$$-\ln\left[\frac{h(t,T)}{h(T,T)}\right] = A(T - t) + \frac{B(T^2 - t^2)}{2}$$

Multiplying both sides by -1 yields

$$\ln\left[\frac{h(t,T)}{h(T,T)}\right] = -\left(A(T - t) + \frac{B(T^2 - t^2)}{2}\right)$$

Exponentiating both sides gives

$$\frac{h(t,T)}{h(T,T)} = \exp\left(-A(T-t) - \frac{B(T^2-t^2)}{2}\right)$$

Multiplying both sides by $h(T,T)$, we obtain

$$h(t,T) = h(T,T)\exp\left\{-\left[A(T-t) + \frac{B(T^2-t^2)}{2}\right]\right\} \tag{21a}$$

Applying the boundary condition (14) on (21a), obtained;

$$h(t,T) = \exp\left\{-[A(T-t) + \frac{B(T^2-t^2)}{2}]\right\}, \tag{21b}$$

from which the optimal value function of the investor is given as;

$$J^*(t,s,w) = \exp\left\{-[A(T-t) + \frac{B(T^2-t^2)}{2}]\right\}\frac{w^{1-\phi}-k}{1-\phi}, \tag{22}$$

which at terminal date, T, equals

$$J^*(T,s,w) = \frac{w^{1-\phi}-k}{1-\phi}. \tag{23}$$

The effects of the parameters d, ϑ, and β are as follows;

When there are no, dividend, (d), transaction costs, (ϑ) , taxes financial market is (β), the optimal investment in the risky asset is,

$$S^*(t) = \frac{[\mu-(\alpha+\sigma t)]}{\emptyset\sigma^2}. \tag{24}$$

The introduction of dividends only gives;

$$S_d{}^*(t) = \frac{[(\mu+d)-(\alpha+\sigma t)]}{\emptyset\sigma^2} \tag{25}$$

Equation (18):

$$S^*_{d,\vartheta,\beta}(t) = \frac{[(\mu+d)-(\alpha+\sigma t+\vartheta+\beta)]}{\phi\sigma^2}w$$

This equation expresses the **dollar amount** invested in the risky asset.

- The term w is the **total wealth** of the investor.
- So $S^*(t)$ here tells you: "How many **dollars** (not fraction) should be allocated to the risky asset at time t?"

Equation (25):

$$S^*_d(t) = \frac{[(\mu+d)-(\alpha+\sigma t)]}{\phi\sigma^2}$$

This equation expresses the **fraction of wealth** allocated to the risky asset.

- The wealth variable w is **not present** here because this is **normalized** investment; i.e., the proportion

$$\frac{\text{dollar amount in risky asset}}{\text{total wealth}}$$

- So this tells you: "What **fraction of wealth** should be in the risky asset if there are only dividends, and no taxes or transaction costs?"

from which,

$$S_d{}^*(t) = \frac{[(\mu + d) - (\alpha + \sigma t)]}{\emptyset\sigma^2}$$

$$= S^*(t) + \frac{d}{\emptyset\sigma^2}. \tag{26}$$

The introduction of transaction costs leads to;

$$S^*_{d,\vartheta,}(t) = \frac{[(\mu + d) - (\alpha + \sigma t + \vartheta)]}{\emptyset\sigma^2}$$

$$= S_d{}^*(t) - \frac{\vartheta}{\emptyset\sigma^2}$$

$$= S^*(t) + \frac{d-\vartheta}{\emptyset\sigma^2}. \tag{27}$$

Introducing tax leads to;

$$S^*_{d,\vartheta,\beta}(t) = \frac{[(\mu + d) - (\alpha + \sigma t + \vartheta + \beta)]}{\emptyset\sigma^2}$$

$$= S_{d,\vartheta}{}^*(t) - \frac{\beta}{\emptyset\sigma^2}$$

$$= S_d{}^*(t) - \frac{(\vartheta + \beta)}{\emptyset\sigma^2}$$

$$= S^*(t) + \frac{d-(\vartheta+\beta)}{\emptyset\sigma^2}. \tag{28}$$

Equation (25) shows that dividends increased the invested amount. Equations (26) through (28) show how the pension investor's investment is depleted by the introduction of each market parameter. Further they show that the investor can only hold investment in the risky asset whenever,

$[(\mu + d) > (\alpha + \sigma t + \vartheta + \beta)]$;

that is, if the sum of the mean return on the risky asset and the dividend ratio is greater than the sum of financial market tax rate, transaction costs, and the rate of return of risk free asset.

The portfolio problem may be described in broad terms as determining the set of decisions to maximize this objective function E [U (.)]. Mathematically this may be written as

max?

$\min_{X_t, C_t, t=0,T-1} \sum_{t=0}^{T} E[U_t(W_t, C_t, R_t, \dots)]$, (29)

where T is the planning horizon, W_t is the wealth at time t, C_t is the consumption at time t, Rt is a vector of returns on the asset classes. Ut (.) is the utility function at time t for Wealth, Return on assets, Consumption or any other properties at each decision point t. The vector X_t is the wealth invested in each asset class at time t; i.e., if there are N asset classes, X_t will be a vector of size N. If instead of a discrete-time problem this were a continuous-time problem, the $\sum U(.)$ would be replaced by an$\int_0^T U(.)$.

Interpreting the Optimization Framework

The general portfolio optimization problem is introduced in abstract terms here:

$$\min_{x_t, c_t} \sum_{t=0}^{T} \mathbb{E}\left[U_t(W_t, C_t, R_t, \dots)\right],$$

where:

- x_t denotes the portfolio weights (fraction of wealth allocated to each asset),
- c_t is the consumption level at time t,
- R_t is a vector of asset returns,
- $U_t(\cdot)$ is the utility function at time t, typically defined over wealth, consumption, and asset returns.

This formulation provides a high-level objective: to choose a sequence of controls $\{x_t, c_t\}$ that maximize expected utility across a finite time horizon. It is a standard representation common to dynamic portfolio choice problems.

However, these symbols are abstract placeholders. In contrast, the specific model developed earlier in the paper describes the investor's wealth process explicitly using the stochastic differential equation:

$$dW^S(t) = S(t)\frac{dP_1(t)}{P_1(t)} + (W(t) - S(t))\frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)S(t)\,dt,$$

where:

- $W(t)$ is total wealth at time t,
- $S(t)$ is the dollar amount invested in the risky asset,
- $P_1(t)$ and $P_0(t)$ are the prices of the risky and risk-free assets, respectively,
- ϑ, β, and d represent transaction costs, tax rate, and dividend rate.

This concrete SDE replaces the abstract terms x_t, R_t, and W_t with economically meaningful quantities that are directly modeled. The goal of the optimization problem becomes one of selecting an investment strategy $S(t)$ that, when plugged into this wealth process, maximizes the expected utility of wealth.

If the optimization problem is one-dimensional, i.e., x $\in \mathbb{R}$, and has no constraints or bounds, a solution$x^*$is characterized by the following conditions, $U^{'}(x^*) = 0$ and $U^{''}(x^*) < 0$, assuming sufficient differentiability of function U. If derivatives of the objective function are available, an approach for finding optima is to find a zero of the function $U^{'}(.)$, which is a necessary condition for optima. Conditions for sufficiency can then be checked at these points. Some standard and well-known methods for finding the zeros of a function are the bisection method, the secant method and Newton's method.

Root-Finding Methods for Solving $U'(x) = 0$

When attempting to find a local optimum of a smooth function $U(x)$, a critical step is identifying where the first derivative vanishes, i.e., solving $U'(x) = 0$. These points are known as *critical points* and may correspond to local minima, maxima, or saddle points.

In many practical scenarios, solving $U'(x) = 0$ analytically is difficult or impossible. In such cases, we rely on **numerical methods** to approximate the solution. Below are three widely used techniques:

Bisection Method:
This is a bracketing method that requires an interval $[a, b]$ such that $U'(a) \cdot U'(b) < 0$. The interval is repeatedly halved until convergence to a root.

- Pros: Guaranteed convergence.
- Cons: Relatively slow; requires sign change.

Secant Method:
This method uses a finite-difference approximation to the derivative and iteratively refines the estimate.

- Pros: Faster than bisection; does not require second derivatives.
- Cons: Convergence is not guaranteed.

Newton's Method:
A fast and widely used method that employs both the first and second derivatives:

$$x_{n+1} = x_n - \frac{U'(x_n)}{U''(x_n)}$$

- Pros: Quadratic convergence near the root.
- Cons: Requires a good initial guess and the existence of $U''(x)$.

These methods are foundational in numerical optimization and play a central role when closed-form solutions are not available.

Conclusion

In this work we have studied the optional investment problem for an investor when transaction costs, dividend (d), and financial market tax rate (β), are involved. We have shown that the introduction of the various market parameters depleted the investment in the risky asset except the dividend which yielded an increment that is a fraction of the wealth.

It is observed that the investor's investment and the value function are horizon dependent, and should be taken into consideration when policy decisions are being made.

It is important to recognize that both the investor's optimal investment strategy and the associated value function are *horizon dependent*. That is, the decisions an investor makes—such as portfolio allocation between risky and risk-free assets—are influenced by the length of the remaining time horizon. For instance, an investor planning for retirement in 30 years will adopt a different strategy than one who is only a few months away from retirement. Likewise, the value function, which quantifies the maximum expected utility or return attainable from a given state and time, evolves as the terminal time approaches. Because both the strategy and its associated value change over time, policy decisions should not be treated as static. Instead, they must explicitly account for the time remaining in the planning horizon to ensure that actions remain optimal throughout the investment period.

If we decide against the linearity of r_t such that the price process of the riskless asset is governed by

$$dP_o = r_t P_o dt \tag{30}$$

where r_t is driven by Ornstein-Uhlenbeck process;

$$dr_t = \alpha(\beta - r_t)dt + \sigma dz(t).\ r_{t_0} = r_0 \tag{31a}$$

or

$$r_t = (r_0 - \beta)e^{-\alpha t} + \sigma \int_{t_0}^{t} e^{-\alpha(t-u)}\, dZ(u), \tag{31b}$$

then, with (2), and using the maximum principle, the Hamilton-Jacobi-Bellman (HJB) equation becomes;

vartheta not sigma

$$J_t + \alpha(\beta - r_t)J_r + [wr_t + [(\mu + d) - (r_t + \sigma + \beta)]s]J_w + r^2 sJ_{rw}$$
$$+\frac{\sigma^2}{2}J_{rr} + \frac{\sigma^2 s^2}{2}J_{ww} = 0 \tag{32}$$

sigma^2 not r^2

Derivation of Equation (32)

We define the value function $J(t, r, W(t))$, where:

- t is time,
- $r(t)$ is the short rate process,
- $W(t)$ is the wealth process,
- $S(t)$ is the control (investment in risky asset).

State Itô's Lemma (Multivariate Form)

Let $J(t, r(t), W(t))$, where both $r(t)$ and $W(t)$ are Itô processes of the form $dX_t = \mu_t\, dt + \sigma_t\, dZ(t)$, driven by the same Brownian motion $Z(t)$. Then:

$$\begin{aligned} dJ &= J_t\, dt + J_r\, dr + J_w\, dW \\ &\quad + \frac{1}{2}J_{rr}\,(dr)^2 + \frac{1}{2}J_{ww}\,(dW)^2 + J_{rw}\, dr\, dW \end{aligned}$$

Model Dynamics

Short Rate Process (Ornstein–Uhlenbeck):

$$dr_t = \alpha(\beta - r_t)\, dt + \sigma\, dZ(t)$$

Wealth Process (corrected form):

$$dW^S(t) = \Big\{W(t)r(t) + [(\mu + d) - r(t) - \vartheta - \beta]\, S(t)\Big\}dt + \sigma S(t)\, dZ(t)$$

Derivation of the Wealth Process with Ornstein–Uhlenbeck Short Rate

We derive the total wealth process $W^S(t)$ under the following assumptions:

- $S(t)$: amount invested in the risky asset,
- $W(t) - S(t)$: amount invested in the risk-free asset,
- The risky asset $P_1(t)$ evolves as:
$$\frac{dP_1(t)}{P_1(t)} = \mu\, dt + \sigma\, dZ_t$$
- The risk-free asset $P_0(t)$ evolves with short rate $r(t)$ as:
$$\frac{dP_0(t)}{P_0(t)} = r(t)\, dt$$
- The short rate $r(t)$ follows an Ornstein–Uhlenbeck process:
$$dr_t = \alpha(\beta - r_t)\, dt + \sigma\, dZ_t$$
- The model includes correction terms for dividends and taxes: ϑ, β, and d.

The total change in wealth is given by the return from both assets and the cost of taxes and dividend leakage:
4
Expanding and simplifying:

$$\begin{aligned} dW^S(t) &= \mu S(t)\, dt + \sigma S(t)\, dZ_t + W(t)r(t)\, dt - S(t)r(t)\, dt - (\vartheta + \beta - d)S(t)\, dt \\ &= [W(t)r(t) + (\mu + d - r(t) - \vartheta - \beta)\, S(t)]\, dt + \sigma S(t)\, dZ_t \end{aligned}$$

Compute Quadratic Variation Terms

We apply Itô's Lemma to the value function $J(t, r_t, W_t)$, where both r_t and W_t are Itô processes driven by the same Brownian motion Z_t.

We use the fundamental stochastic calculus identities:

$$(dZ_t)^2 = dt, \quad dt \cdot dZ_t = 0, \quad (dt)^2 = 0$$

Short Rate Process:

The short rate process follows:
$$dr_t = \alpha(\beta - r_t)\, dt + \sigma\, dZ_t$$

We compute:
$$\begin{aligned} (dr_t)^2 &= [\alpha(\beta - r_t)\, dt + \sigma\, dZ_t]^2 \\ &= \alpha^2(\beta - r_t)^2\, (dt)^2 + 2\alpha(\beta - r_t)\sigma\, dt\, dZ_t + \sigma^2\, (dZ_t)^2 \\ &= 0 + 0 + \sigma^2\, dt \\ &= \sigma^2\, dt \end{aligned}$$

Wealth Process:

I'll explain what that equation is saying in plain English, then show how to solve it to get the bond (money-market) price path, and what "short rate" means economically.

That line is just the **continuous-time version of "your bank account earns interest at rate $r(t)$"**.

What the equation says

$$\frac{dP_0(t)}{P_0(t)} = r(t)\,dt$$

- $P_0(t)$: value of the **risk-free asset** (think: money-market account / bank account / continuously-rolling T-bill).
- $r(t)$: the **short rate** = the *instantaneous* risk-free interest rate at time t.
- $\frac{dP_0(t)}{P_0(t)}$: the **instantaneous proportional (percentage) change** in P_0.

So it's equivalent to:

$$dP_0(t) = r(t)\,P_0(t)\,dt$$

Meaning: over a tiny time step dt, the asset increases by "rate × current value × time".

Tiny-time intuition

For a small Δt,

$$P_0(t+\Delta t) \approx P_0(t)\big(1 + r(t)\Delta t\big)$$

That's the usual "simple interest" approximation over a short interval.

Solving it (the closed form)

Divide both sides by $P_0(t)$ and integrate:

$$\frac{dP_0(t)}{P_0(t)} = r(t)\,dt \quad \Rightarrow \quad d(\ln P_0(t)) = r(t)\,dt$$

Integrate from 0 to t:

$$\ln P_0(t) - \ln P_0(0) = \int_0^t r(s)\,ds$$

Exponentiate:

$$\boxed{P_0(t) = P_0(0)\exp\left(\int_0^t r(s)\,ds\right)}$$

Special case: constant rate

If $r(t) = r$ is constant:

$$\boxed{P_0(t) = P_0(0)e^{rt}}$$

which is exactly **continuous compounding**.

Why there's no randomness term

In many models, risky assets have something like $+\sigma\,dW_t$. Here there's **no** dW_t term because it's defined to be **risk-free**: over an instant, it only earns interest, no uncertainty.

If you want, paste the surrounding lines (or the definition of $P_1(t)$ above it), and I'll explain how this ties into the risky asset and the wealth equation.

The wealth process is given by:

$$dW_t = [W_t r_t + ((\mu + d) - r_t - \vartheta - \beta)\, S_t]\, dt + \sigma S_t\, dZ_t$$

Let us define:

$$A_t := W_t r_t + ((\mu + d) - r_t - \vartheta - \beta)\, S_t$$

Then:

$$dW_t = A_t\, dt + \sigma S_t\, dZ_t$$

We compute:

$$\begin{aligned}(dW_t)^2 &= [A_t\, dt + \sigma S_t\, dZ_t]^2 \\ &= A_t^2\,(dt)^2 + 2A_t \sigma S_t\, dt\, dZ_t + \sigma^2 S_t^2\,(dZ_t)^2 \\ &= 0 + 0 + \sigma^2 S_t^2\, dt \\ &= \sigma^2 S_t^2\, dt\end{aligned}$$

Cross-Term: $dr_t \cdot dW_t$

We compute:

$$\begin{aligned}dr_t \cdot dW_t &= [\alpha(\beta - r_t)\, dt + \sigma\, dZ_t] \cdot [A_t\, dt + \sigma S_t\, dZ_t] \\ &= \alpha(\beta - r_t) A_t\,(dt)^2 + \alpha(\beta - r_t)\sigma S_t\, dt\, dZ_t + \sigma A_t\, dZ_t\, dt + \sigma^2 S_t\,(dZ_t)^2 \\ &= 0 + 0 + 0 + \sigma^2 S_t\, dt \\ &= \sigma^2 S_t\, dt\end{aligned}$$

Final Summary:

$$\begin{aligned}(dr_t)^2 &= \sigma^2\, dt \\ (dW_t)^2 &= \sigma^2 S_t^2\, dt \\ dr_t \cdot dW_t &= \sigma^2 S_t\, dt\end{aligned}$$

These second-order terms are required when applying Itô's Lemma to $J(t, r_t, W_t)$, and appear in the final HJB equation through the terms:

$$\frac{1}{2} J_{rr} (dr_t)^2, \quad \frac{1}{2} J_{ww} (dW_t)^2, \quad J_{rw}\, dr_t \cdot dW_t$$

Apply Itô's Lemma

Substitute into Itô's Lemma:

$$\begin{aligned}dJ = J_t\, dt &+ J_r\, [\alpha(\beta - r(t))\, dt + \sigma\, dZ(t)] \\ &+ J_w\, \{[W r(t) + ((\mu + d) - r(t) - \vartheta - \beta)\, S(t)]\, dt + \sigma S(t)\, dZ(t)\} \\ &+ \frac{1}{2} J_{rr}\, \sigma^2\, dt + \frac{1}{2} J_{ww}\, \sigma^2 S(t)^2\, dt + J_{rw}\, \sigma^2 S(t)\, dt\end{aligned}$$

Extract Drift Terms for the HJB

The HJB uses only drift terms (the dt-terms). Group them:

$$\begin{aligned}0 = J_t &+ \alpha(\beta - r(t))\, J_r \\ &+ [W r(t) + ((\mu + d) - r(t) - \vartheta - \beta)\, S(t)]\, J_w \\ &+ \frac{1}{2}\sigma^2\, J_{rr} + \frac{1}{2}\sigma^2 S(t)^2\, J_{ww} + \sigma^2 S(t)\, J_{rw}\end{aligned}$$

This is the Hamilton-Jacobi-Bellman (HJB) equation corresponding to the wealth and short rate dynamics under control $S(t)$.

Yes — in your exact context:

$$(dr_t)^2 = \sigma^2 dt \quad \text{and} \quad (dr)^2 = \sigma^2 dt \quad \Longrightarrow \quad (dr_t)^2 = (dr)^2$$

Differentiating (32) with respect to s gives

$$[(\mu + d) - (r_t + \vartheta + \beta)]J_w + \sigma^2 J_{rw} + \sigma^2 s J_{ww} \tag{33}$$

from which

$$S^* = -\frac{[(\mu+d)-(r_t+\vartheta+\beta)]J_w}{\sigma^2 J_{ww}} - \frac{\sigma^2 J_{rw}}{\sigma^2 J_{ww}}. \tag{34}$$

To eliminate dependency on w are conjecture that

$$J(t,r,w) = G(t,r)\frac{[w^{1-\emptyset}]}{1-\phi}; \phi \neq 1 \tag{35a}$$

with the boundary condition

$$G(T,r) = 1. \tag{35b}$$

From (35a) obtain

$$J_w = w^{-\phi}G,\ J_{ww} = -\phi w^{-\phi-1} G and J_{rw} = w^{-\phi} G_r \tag{35c}$$

Applying (35c) to (34) yields

$$S^*{}_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{[(\mu+d)-(r_t+\vartheta+\beta)]}{\sigma^2} + \frac{G_r}{G}\right]. \tag{36}$$

Substituting (35c) into (34)

We begin with equation (34), which is:

$$S^* = -\frac{[(\mu + d) - (r_t + \vartheta + \beta)]J_w}{\sigma^2 J_{ww}} - \frac{J_{rw}}{J_{ww}}$$

From equation (35c), we know the following expressions for the derivatives of $J(t, r, w)$:

$$J_w = w^{-\phi}G,$$
$$J_{ww} = -\phi w^{-\phi-1}G,$$
$$J_{rw} = w^{-\phi}G_r$$

Substitute these into equation (34):

$$S^* = -\frac{[(\mu + d) - (r_t + \vartheta + \beta)] \cdot w^{-\phi}G}{\sigma^2 \cdot (-\phi w^{-\phi-1}G)} - \frac{w^{-\phi}G_r}{-\phi w^{-\phi-1}G}$$

Now simplify each term carefully:

First Term:

$$-\frac{[(\mu+d)-(r_t+\vartheta+\beta)]\cdot w^{-\phi}G}{\sigma^2\cdot(-\phi w^{-\phi-1}G)} = \frac{[(\mu+d)-(r_t+\vartheta+\beta)]\cdot w^{-\phi}G}{\phi\sigma^2 w^{-\phi-1}G}$$
$$= \frac{[(\mu+d)-(r_t+\vartheta+\beta)]\cdot w^{-\phi}G}{\phi\sigma^2\cdot w^{-\phi-1}G}$$
$$= \frac{[(\mu+d)-(r_t+\vartheta+\beta)]}{\phi\sigma^2}\cdot w$$

Second Term:

$$-\frac{w^{-\phi}G_r}{-\phi w^{-\phi-1}G} = \frac{w^{-\phi}G_r}{\phi w^{-\phi-1}G}$$
$$= \frac{w^{-\phi}G_r}{\phi\cdot w^{-\phi-1}G}$$
$$= \frac{G_r}{\phi G}\cdot w$$

Final Combined Expression:

$$S^* = \frac{w}{\phi}\left[\frac{(\mu+d)-(r_t+\vartheta+\beta)}{\sigma^2} - \frac{G_r}{G}\right]$$

This is the closed-form optimal strategy as a function of current wealth w, short rate r_t, and the gradient of the reduced value function G.

Further to eliminate dependency on$r$ we conjecture that

$$G(t,r) = h(t)\left[\frac{r^{1-\phi}}{1-\phi}\right], \tag{37a}$$

such that

$$h(T) = \frac{1-\phi}{r^{1-\phi}}\,. \tag{37b}$$

We obtain from (37a)

$$G_r = r^{-\phi}h(t). \tag{37c}$$

The application of (37a) and (37c) in (36) yields

$$S^*{}_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{[(\mu+d)-(r_t+\vartheta+\beta)]}{\sigma^2} + (1-\phi)r_t\right] \quad \text{mistake?} \tag{38}$$

Derivation of $\frac{G_r}{G} = \frac{1-\phi}{r}$

We are given the conjectured functional form (Equation 37a):

$$G(t,r) = h(t)\left(\frac{r^{1-\phi}}{1-\phi}\right)$$

This is a separable function in t and r, with all r-dependence isolated in the power term.

Step 1: Differentiate $G(t,r)$ with respect to r

We compute the partial derivative G_r by differentiating only the r-dependent component:

$$G_r(t,r) = h(t) \cdot \frac{d}{dr}\left(\frac{r^{1-\phi}}{1-\phi}\right) = h(t) \cdot r^{-\phi}$$

Step 2: Divide by $G(t,r)$

Recall:

$$G(t,r) = h(t) \cdot \frac{r^{1-\phi}}{1-\phi}$$

Therefore, the ratio becomes:

$$\frac{G_r}{G} = \frac{h(t) \cdot r^{-\phi}}{h(t) \cdot \frac{r^{1-\phi}}{1-\phi}} = \frac{(1-\phi) \cdot r^{-\phi}}{r^{1-\phi}} = \frac{1-\phi}{r}$$

Conclusion

The final expression is:

$$\frac{G_r}{G} = \frac{1-\phi}{r}$$

Note that this is **not** equal to $(1-\phi)r$, unless one mistakenly assumes $r = \frac{1}{r}$. If the original text claims $\frac{G_r}{G} = (1-\phi)r$, then either there is a notational clash, or the result is in error. The correct derivation shows that:

$$\frac{G_r}{G} \text{ is inversely proportional to } r$$

under the given conjectured form for $G(t,r)$.

Notice that the optimal investment in the risky asset, using (31b), will become

$$S^*{}_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{[(\mu+d)-(\vartheta+\beta)]}{\sigma^2} + \left(1-\frac{1}{\sigma^2}-\phi\right)\left[(r_0-\beta)e^{-\propto t} + \sigma\int_{t_0}^{t} e^{-\propto(t-u)}dZ(u)\right]\right]. \quad (39)$$

Notice that

I believe equation 31a is wrong
look at Miscellaneous

$$\lim_{t-\infty} S^*{}_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{[(\mu+d)-(\vartheta+\beta)]}{\sigma^2}\right], \quad (40)$$

while

This has measure as T ---> t_0?

$$\lim_{t-0} S^*{}_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{[(\mu+d)-(\vartheta+\beta)]}{\sigma^2} + \left(1-\frac{1}{\sigma^2}-\phi\right)\left[(r_0-\beta) + \sigma\int_{t_0}^{T} e^{\propto u}dZ(u)\right]\right]. \quad (41)$$

Corrected Derivation of Equation (39)

We begin with the corrected expression from Equation (38):

$$S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{(\mu+d)-(r_t+\vartheta+\beta)}{\sigma^2} + (1-\phi)r_t\right] \quad (38)$$

Group the r_t terms:

$$S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{(\mu+d)-(\vartheta+\beta)}{\sigma^2} + r_t\left(1-\phi-\frac{1}{\sigma^2}\right)\right]$$

Next, substitute the known solution for the short rate r_t under the Ornstein–Uhlenbeck process:

$$r_t = \beta + (r_0 - \beta)e^{-\alpha t} + \sigma \int_{t_0}^{t} e^{-\alpha(t-u)}\, dZ(u)$$

Plugging this into the expression above yields:

$$S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{(\mu+d)-(\vartheta+\beta)}{\sigma^2} + \left(1-\phi-\frac{1}{\sigma^2}\right)\left(\beta + (r_0-\beta)e^{-\alpha t} + \sigma\int_{t_0}^{t} e^{-\alpha(t-u)}\, dZ(u)\right)\right]$$

Distribute the coefficient:

$$\begin{aligned} S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\Bigg[&\frac{(\mu+d)-(\vartheta+\beta)}{\sigma^2} + \left(1-\phi-\frac{1}{\sigma^2}\right)\beta + \left(1-\phi-\frac{1}{\sigma^2}\right)(r_0-\beta)e^{-\alpha t} \\ &+ \left(1-\phi-\frac{1}{\sigma^2}\right)\sigma\int_{t_0}^{t} e^{-\alpha(t-u)}\, dZ(u)\Bigg] \end{aligned}$$

Limits of the Optimal Investment Strategy

We begin with the expanded expression:

$$\begin{aligned} S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\Bigg[&\frac{(\mu+d)-(\vartheta+\beta)}{\sigma^2} + \left(1-\phi-\frac{1}{\sigma^2}\right)\beta + \left(1-\phi-\frac{1}{\sigma^2}\right)(r_0-\beta)e^{-\alpha t} \\ &+ \left(1-\phi-\frac{1}{\sigma^2}\right)\sigma\int_{t_0}^{t} e^{-\alpha(t-u)} dZ(u)\Bigg] \end{aligned}$$

Limit as $t \to \infty$

As $t \to \infty$, we have:

$$e^{-\alpha t} \to 0, \qquad \int_{t_0}^{t} e^{-\alpha(t-u)} dZ(u) \to 0 \text{ (by the zero mean property).}$$

Therefore:

$$\lim_{t\to\infty} S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{(\mu+d)-(\vartheta+\beta)}{\sigma^2} + \left(1-\phi-\frac{1}{\sigma^2}\right)\beta\right]$$

Limit as $t \to 0$

At $t = t_0$, we have:

$$e^{-\alpha t} \to 1, \qquad \int_{t_0}^{t} e^{-\alpha(t-u)} dZ(u) \to 0.$$

So:

$$\lim_{t\to t_0} S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{(\mu+d)-(\vartheta+\beta)}{\sigma^2} + \left(1-\phi-\frac{1}{\sigma^2}\right)\beta + \left(1-\phi-\frac{1}{\sigma^2}\right)(r_0-\beta)\right]$$

Miscellaneous

How the Utility Function Encodes Risk Aversion

Even though a concave utility function assigns higher utility to higher wealth, it does **not** imply that the agent simply maximizes wealth. Instead, the **curvature** of the utility function captures *risk aversion.*

Numerical Example

Suppose an investor faces two strategies:

- Strategy A: 50% chance of ending with \$100, 50% chance of ending with \$0.
- Strategy B: guaranteed \$50.

Both have the same expected wealth:

$$\mathbb{E}[W] = 0.5 \cdot 100 + 0.5 \cdot 0 = 50$$

But under log utility:

$$\mathbb{E}[\log(W)] = 0.5 \cdot \log(100) + 0.5 \cdot \log(0) = -\infty$$
$$\log(50) \approx 3.91$$

So even though the expected wealth is identical, the **expected utility is dramatically lower** for the risky strategy.

Key Check: Does the Solution Mean-Revert to β?

We begin with the Ornstein–Uhlenbeck process:

$$dr_t = \alpha(\beta - r_t)\, dt + \sigma\, dZ_t, \quad r_{t_0} = r_0$$

The solution to this SDE is:

$$r_t = (r_0 - \beta)e^{-\alpha(t-t_0)} + \beta + \sigma \int_{t_0}^{t} e^{-\alpha(t-u)}\, dZ(u)$$

We now verify that this solution satisfies the key property of mean reversion toward β by computing its expectation.

Taking Expectation

Using linearity of expectation:

$$\begin{aligned}\mathbb{E}[r_t] &= \mathbb{E}\left[(r_0 - \beta)e^{-\alpha(t-t_0)}\right] + \mathbb{E}[\beta] + \mathbb{E}\left[\sigma \int_{t_0}^{t} e^{-\alpha(t-u)}\, dZ(u)\right] \\ &= (r_0 - \beta)e^{-\alpha(t-t_0)} + \beta + 0\end{aligned}$$

The final term vanishes because the Itô integral has zero mean:

$$\mathbb{E}\left[\int_{t_0}^{t} e^{-\alpha(t-u)}\, dZ(u)\right] = 0$$

Zero mean property:

$$\mathbb{E}\left[\int_{t_0}^{t} f(u)\, dZ(u)\right] = 0 \qquad \text{(if } f \text{ is deterministic)}$$

Long-Run Limit

Taking the limit as $t \to \infty$:

$$\lim_{t\to\infty} \mathbb{E}[r_t] = \lim_{t\to\infty} \left[(r_0 - \beta)e^{-\alpha(t-t_0)} + \beta\right] = \beta$$

Conclusion

This confirms that the solution correctly reflects mean-reversion toward β, as defined by the dynamics of the SDE:

$$dr_t = \alpha(\beta - r_t)\,dt + \sigma\,dZ_t$$

If the solution omitted the β term, we would instead have:

$$\lim_{t\to\infty} \mathbb{E}[r_t] = 0$$

which contradicts the entire structure of the model. Therefore, the inclusion of the β term is essential for the solution to be valid.

What Happens to a Definite Integral as the Interval Collapses?

You're asking: what happens to the definite integral

$$\int_a^b f(x)\,dx$$

as $b \to a$?

Answer

If f is integrable on $[a, b]$, then as $b \to a$, the length of the interval shrinks to zero. Therefore:

$$\lim_{b\to a} \int_a^b f(x)\,dx = 0$$

This is true because you're integrating over a region with vanishing measure. The "area under the curve" collapses to zero width.

Example

Let $f(x) = x^2$. Then:

$$\int_a^b x^2\,dx = \left[\frac{1}{3}x^3\right]_a^b = \frac{1}{3}(b^3 - a^3)$$

Taking the limit:

$$\lim_{b\to a} \int_a^b x^2\,dx = \lim_{b\to a} \frac{1}{3}(b^3 - a^3) = 0$$

When Can You Plug In to Evaluate a Limit?

If the function inside the limit is continuous at the point being approached, then you can directly evaluate the limit by substitution:

$$\lim_{x\to a} f(x) = f(a) \quad \text{(if } f \text{ is continuous at } a\text{).}$$

However, if the function results in an indeterminate form (like $0/0$, ∞/∞, etc.), then you **cannot** just plug in. In such cases, you must use algebraic simplification, L'Hôpital's Rule, or another technique to evaluate the limit properly.

Terminal Condition for the Value Function in an HJB Problem

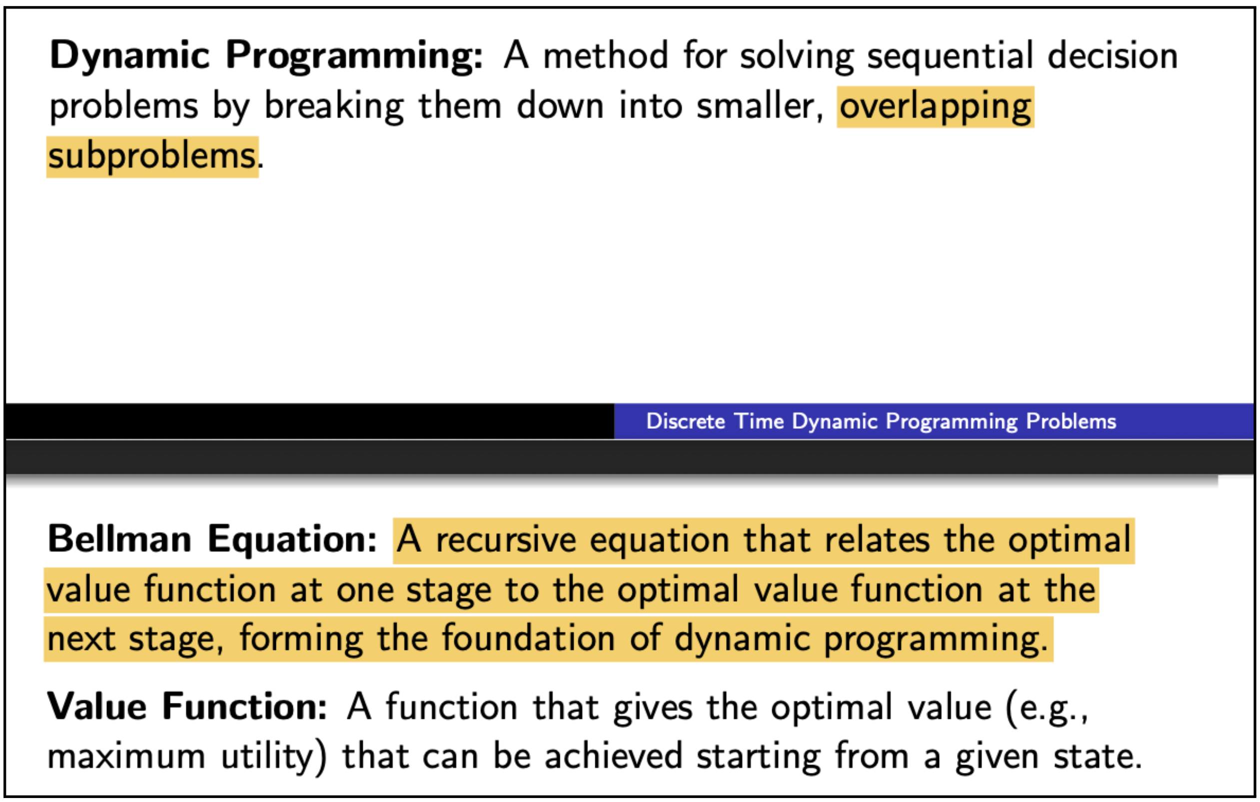
The terminal condition for the value function arises naturally in dynamic optimization problems.

- $V(t, w)$ is the *maximum expected utility of wealth* starting from time t with wealth w.
- At the final time T, there are no future decisions — the agent simply consumes or realizes their wealth.
- Therefore, the best the agent can do at T is to obtain utility from their current wealth:

$$V(T, w) = U(w).$$

- This serves as the *boundary condition* when solving the HJB PDE backward in time.

Discrete DP Notes



Talking: Ogbu Imaga

1.2 The principle of optimality

A key idea is that optimization over time can often be regarded as 'optimization in stages'. We trade off our desire to obtain the lowest possible cost at the present stage against the implication this would have for costs at future stages. The best action minimizes the sum of the cost incurred at the current stage and the least total cost that can be incurred from all subsequent stages, consequent on this decision. This is known as the Principle of Optimality.

Definition 1.1 (Principle of Optimality) From any point on an optimal trajectory, the remaining trajectory is optimal for the corresponding problem initiated at that point.

If a path is optimal from the start to the finish, then any part of that path—from any point onward—is also optimal for the smaller problem that starts at that point.

Discrete DP Example

Maximize the total reward:

$$\max_{u_0,u_1,u_2,u_3} \sum_{t=0}^{3} \left(1 + x_t - u_t^2\right)$$

subject to:

$$x_{t+1} = x_t + u_t, \quad x_0 = 0, \quad u_t \in \mathbb{R}.$$

Understand the Structure

This is a finite-horizon deterministic control problem with:

$$\text{State: } x_t, \quad \text{Control: } u_t, \quad \text{Dynamics: } x_{t+1} = x_t + u_t, \quad \text{Reward: } f(t, x_t, u_t) = 1 + x_t - u_t^2.$$

We use **Bellman recursion**:

$$J_t(x) = \max_{u \in \mathbb{R}} \left[1 + x - u^2 + J_{t+1}(x+u)\right].$$

Terminal Time $t = 3$

At final time:

$$J_3(x) = \max_{u \in \mathbb{R}} \left[1 + x - u^2\right].$$

Differentiate:

$$\frac{d}{du}(1 + x - u^2) = -2u = 0 \quad \Rightarrow \quad u = 0.$$

Thus:

$$J_3(x) = 1 + x.$$

Time $t = 2$

$$J_2(x) = \max_u \left[1 + x - u^2 + J_3(x+u)\right] = \max_u \left[1 + x - u^2 + 1 + (x+u)\right].$$

Simplify:

$$J_2(x) = \max_u \left[2 + 2x + u - u^2\right].$$

Differentiate:

$$1 - 2u = 0 \quad \Rightarrow \quad u = \frac{1}{2}.$$

Plug back:

$$J_2(x) = 2 + 2x + \frac{1}{2} - \frac{1}{4} = 2.25 + 2x.$$

Time $t = 1$

$$J_1(x) = \max_u \left[1 + x - u^2 + J_2(x+u)\right] = \max_u \left[1 + x - u^2 + 2.25 + 2(x+u)\right].$$

Simplify:

$$J_1(x) = \max_u \left[3.25 + 3x + 2u - u^2\right].$$

Differentiate:

$$2 - 2u = 0 \quad \Rightarrow \quad u = 1.$$

Plug back:

$$J_1(x) = 3.25 + 3x + 2(1) - 1 = 4.25 + 3x.$$

Time $t = 0$, $x_0 = 0$

$$J_0(0) = \max_u \left[1 - u^2 + J_1(u)\right] = \max_u \left[1 - u^2 + 4.25 + 3u\right].$$

Simplify:

$$J_0(0) = \max_u \left[5.25 + 3u - u^2\right].$$

Differentiate:

$$3 - 2u = 0 \quad \Rightarrow \quad u = \frac{3}{2}.$$

Plug back:

$$J_0(0) = 5.25 + \frac{9}{2} - \frac{9}{4} = 5.25 + 4.5 - 2.25 = 7.5.$$

Final Answer

Maximum reward:

$$7.5$$

Optimal controls:

$$u_0 = \frac{3}{2}, \quad u_1 = 1, \quad u_2 = \frac{1}{2}, \quad u_3 = 0.$$

States:

$$x_0 = 0, \quad x_1 = \frac{3}{2}, \quad x_2 = \frac{5}{2}, \quad x_3 = 3, \quad x_4 = 3.$$

bellman equation uses bellman principal?

Yes — the **Bellman equation** is essentially the mathematical expression of the **Bellman Principle of Optimality**.

🔷 **Step 2**

You didn't guess u_2.
You assumed:

> "If I pick control u_2, the state will become $x_3 = x_2 + u_2$,
> and **from that point on, I'll behave optimally** using $J_3(x_3)$."

So you used:

$$J_2(x_2) = \max_{u_2} \left[1 + x_2 - u_2^2 + J_3(x_2 + u_2)\right]$$

That's Bellman's Principle. You **optimize right now**, assuming **optimal behavior in the future**.

Continuous DP Notes

Hamilton-Jacobi-Bellman (HJB) Equation: A partial differential equation that the optimal value function satisfies. It represents the dynamic programming principle in continuous time.
Value Function (V(t, x)): The optimal value of the objective function (e.g., total reward) from time *t* onwards, given the state x at time t. The best possible value you can achieve for your goal, starting from time t in state x, assuming you act optimally from that point on.

A continuous-time optimal control problem involves finding an optimal control function within a continuous time interval that minimizes (or maximizes) a given cost function while satisfying specified constraints on the system's dynamics.

1. **Define the Problem:** Clearly state the objective (e.g., maximize total reward, minimize cost), the state and control variables, the dynamic system, the instantaneous reward function, and any boundary or terminal conditions.
2. **Derive the HJB Equation:** Apply Itô's formula (for stochastic problems) to the value function and use the dynamic programming principle to obtain the HJB equation. Talking about approach in paper not in the slides
3. **Solve the HJB Equation:** Solve the HJB equation, often using techniques like backward induction or approximation methods, to find the optimal value function.
4. **Determine the Optimal Policy:** Use the optimal value function to find the optimal control policy ($u^*(t, x)$) that maximizes the value function at each time and state.
5. **Verification (Optional):** Verify that the solution obtained satisfies the original problem's conditions and the HJB equation.

Dynamic Programming Solution via HJB Equation

System Dynamics:

$$\dot{x}(t) = -2x(t) + u(t), \quad x(0) = x_0$$

Objective Functional:

$$J = \int_0^\infty \left(-x(t)^2 - u(t)^2\right)\, dt$$

We aim to determine the optimal control $u(t)$ that maximizes the objective functional.

Step 1: Value Function Ansatz

We assume the value function takes the form:

$$V(x) = -Px^2, \quad P > 0$$

which is a quadratic function suitable for linear-quadratic control problems with maximization.

Step 2: Hamilton-Jacobi-Bellman (HJB) Equation

For an infinite horizon, the HJB equation is:

undiscounted infinite horizon

$$0 = \max_u \left\{-x^2 - u^2 + V'(x)(-2x + u)\right\}$$

Refer to slide 7

Since $V(x) = -Px^2$, we compute:

$$V'(x) = -2Px$$

Substitute into the HJB:

$$\begin{aligned} 0 &= \max_u \left\{-x^2 - u^2 + (-2Px)(-2x + u)\right\} \\ &= \max_u \left\{-x^2 - u^2 + 4Px^2 - 2Pxu\right\} \end{aligned}$$

Step 3: Optimal Control

To maximize, take the derivative with respect to u:

$$\frac{d}{du}\left(-x^2 - u^2 + 4Px^2 - 2Pxu\right) = -2u - 2Px = 0 \Rightarrow u^* = -Px$$

Step 4: Plug Optimal Control into HJB
Substitute $u = -Px$ into the HJB:

$$\begin{aligned} 0 &= -x^2 - (Px)^2 + 4Px^2 - 2Px(-Px) \\ &= -x^2 - P^2x^2 + 4Px^2 + 2P^2x^2 \\ &= x^2\left(-1 + 4P + P^2\right) \end{aligned}$$

Step 5: Solve Algebraic Riccati Equation
We solve:

$$-1 + 4P + P^2 = 0 \Rightarrow P^2 + 4P - 1 = 0$$

Solving this quadratic equation:

$$P = \frac{-4 \pm \sqrt{16+4}}{2} = \frac{-4 \pm \sqrt{20}}{2} = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}$$

We choose the positive root:

$$P = -2 + \sqrt{5}$$

Final Result

Optimal Control:

$$u^*(t) = -Px(t) = -(-2 + \sqrt{5})x(t)$$

Value Function:

$$V(x) = -Px^2 = -(-2 + \sqrt{5})x^2$$